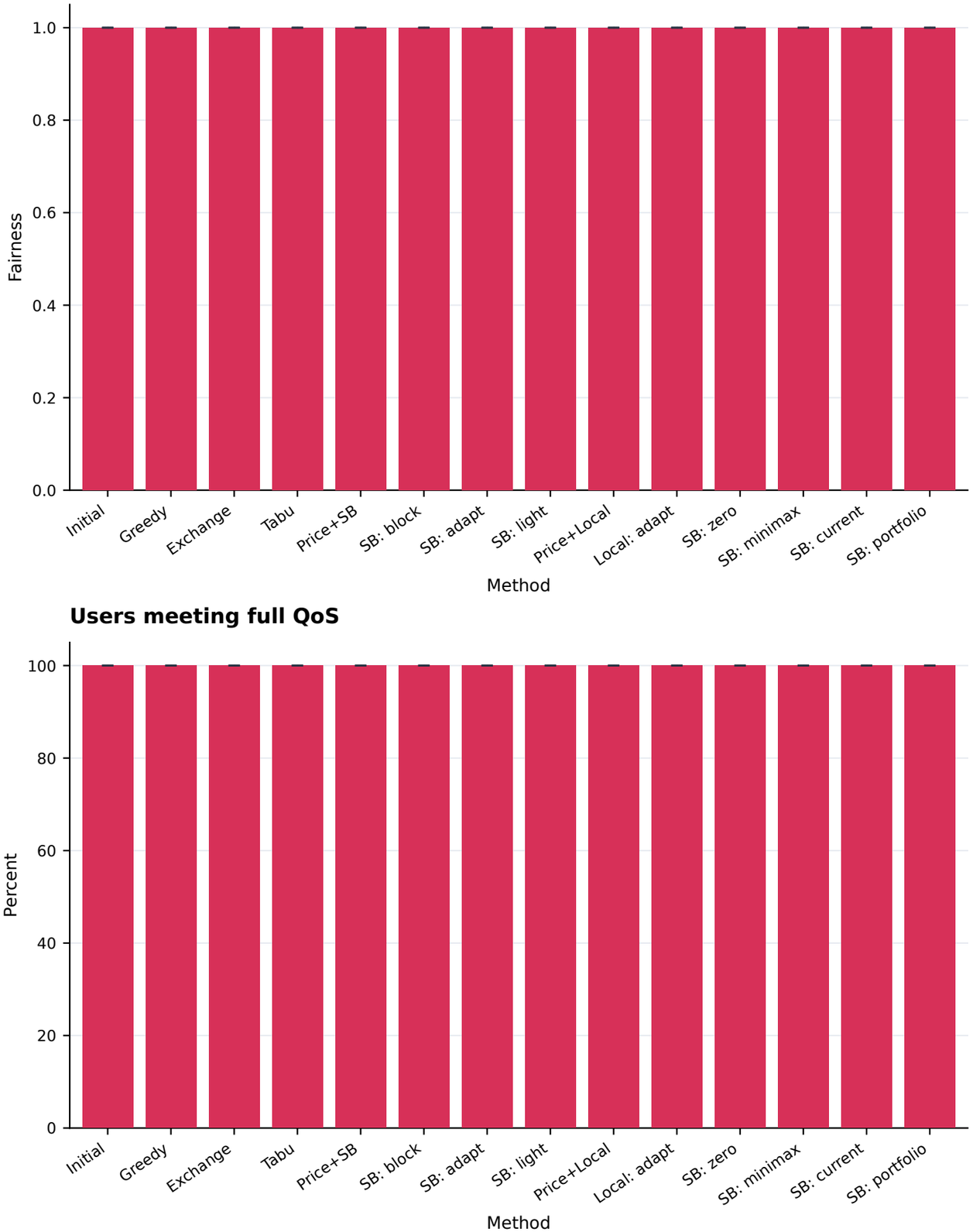
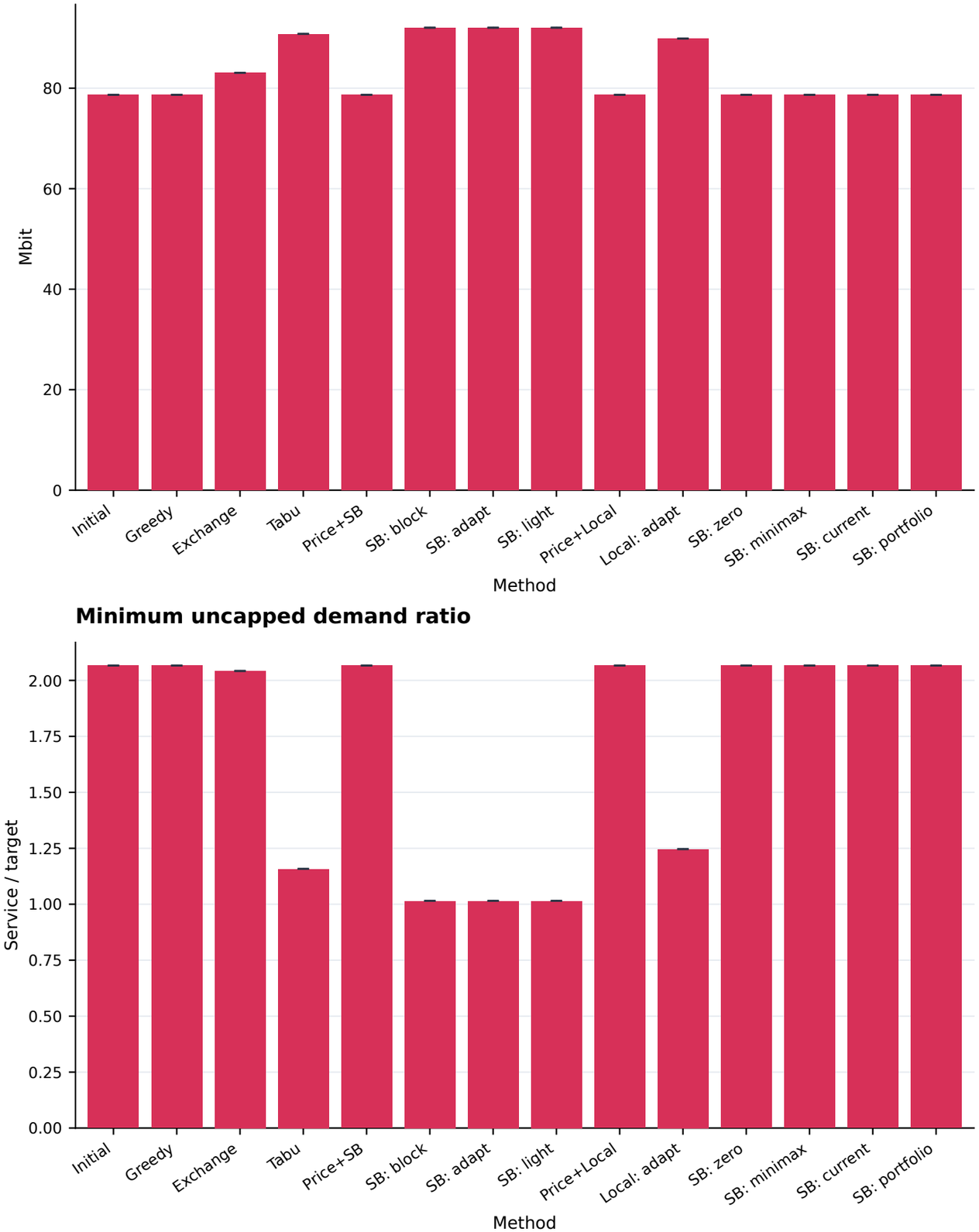
custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

Capped minimum QoS ratio



Certified service ONLY for full-QoS schedules

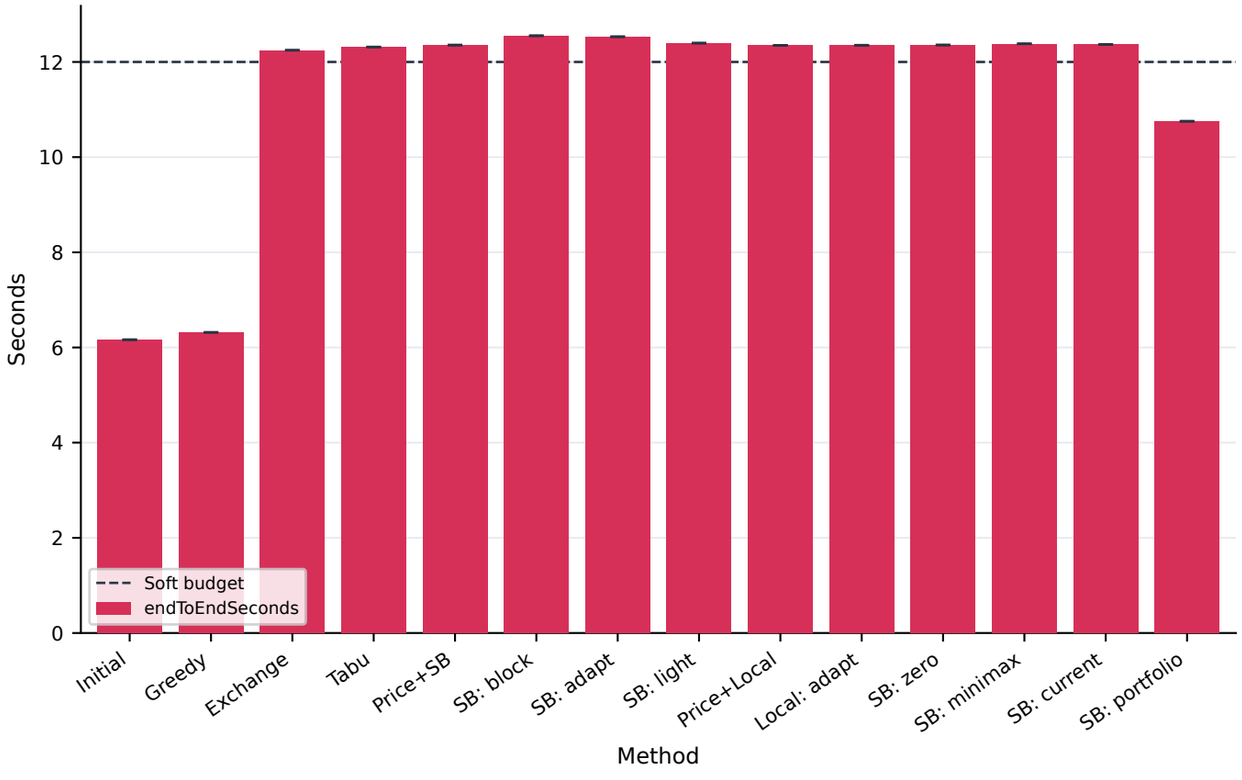


Users meeting full QoS

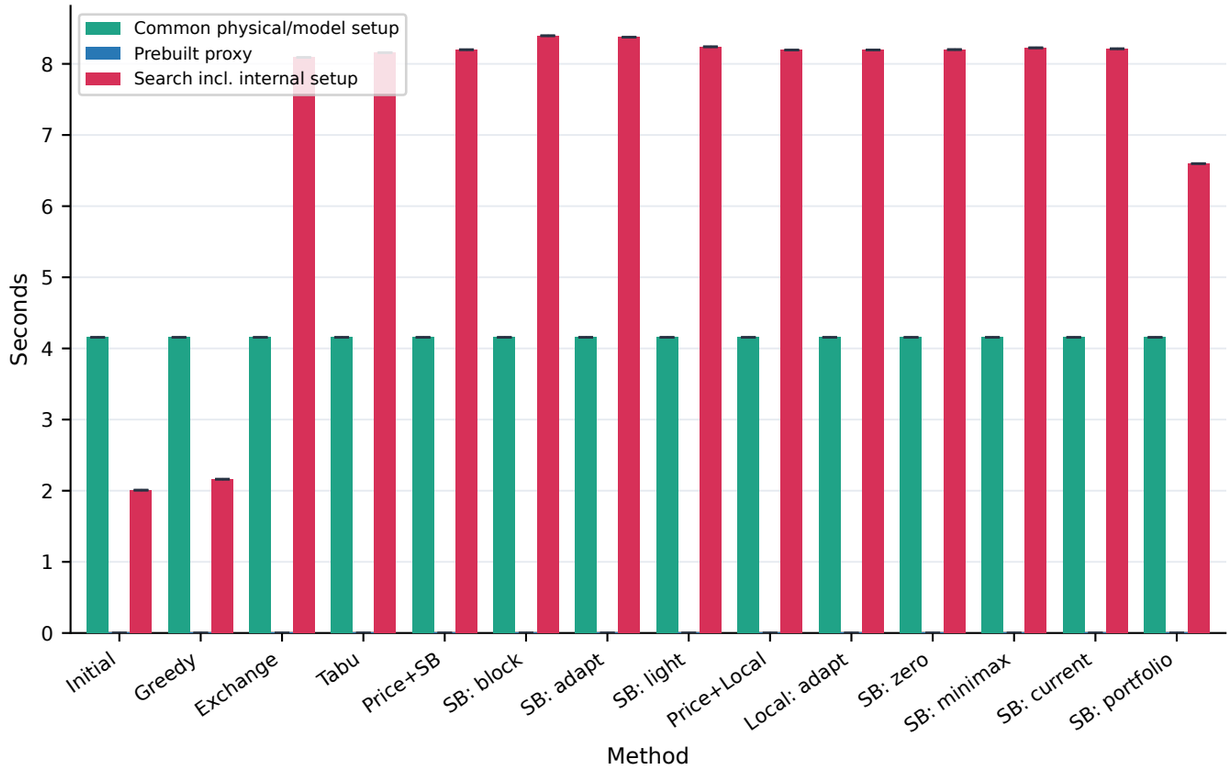
Minimum uncapped demand ratio

custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

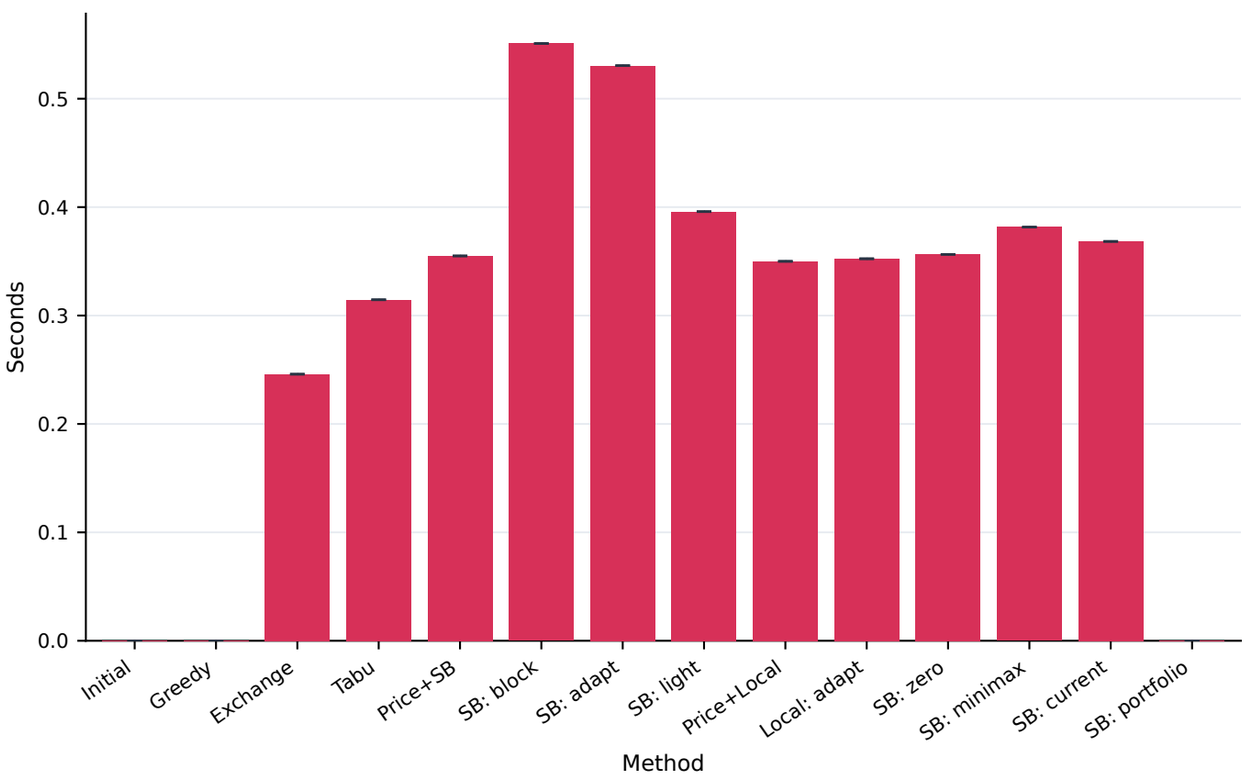
Accounted end-to-end time



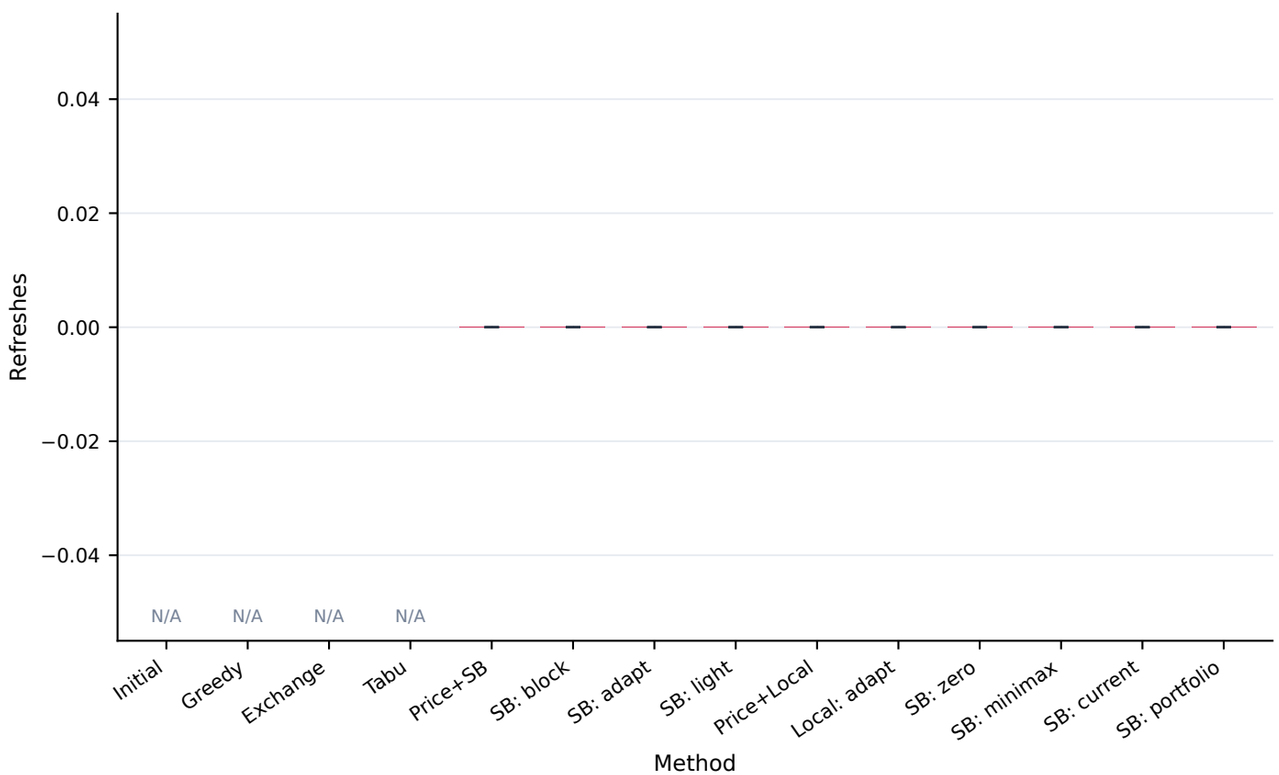
Timing components (internal proxy work stays in search)



Actual budget overrun (not clipped)

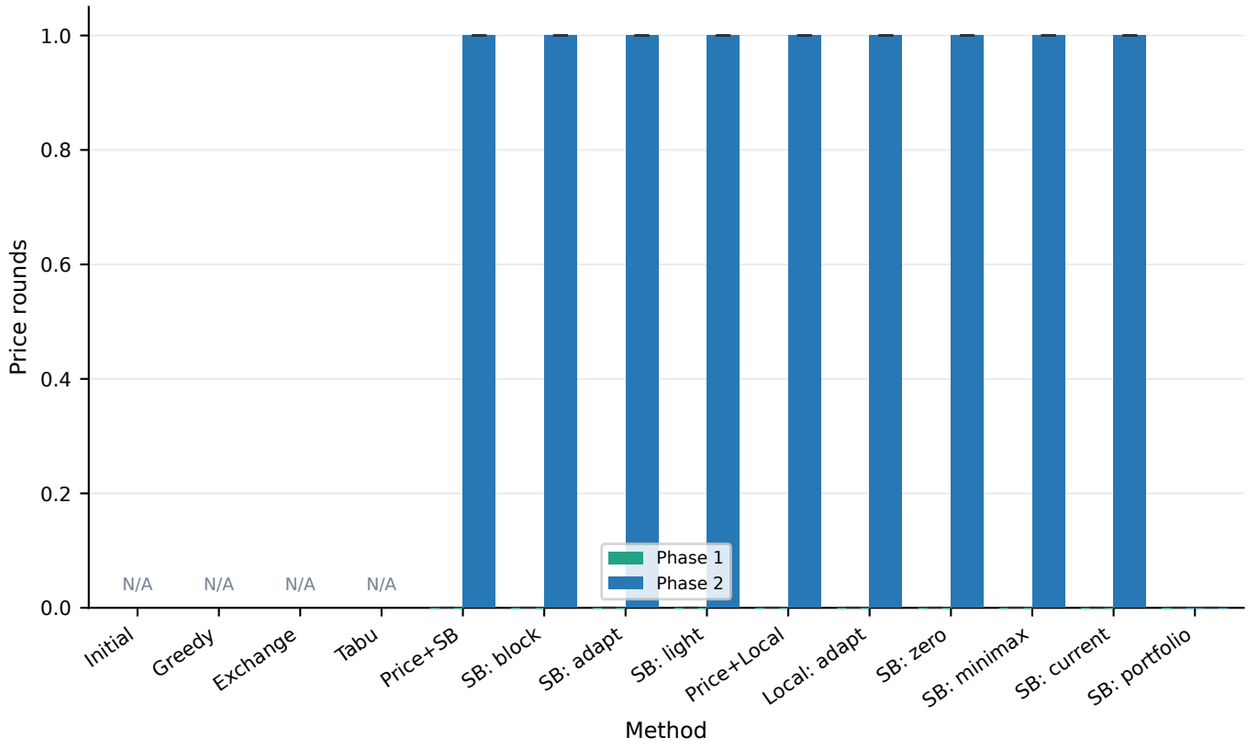


Proxy refresh count

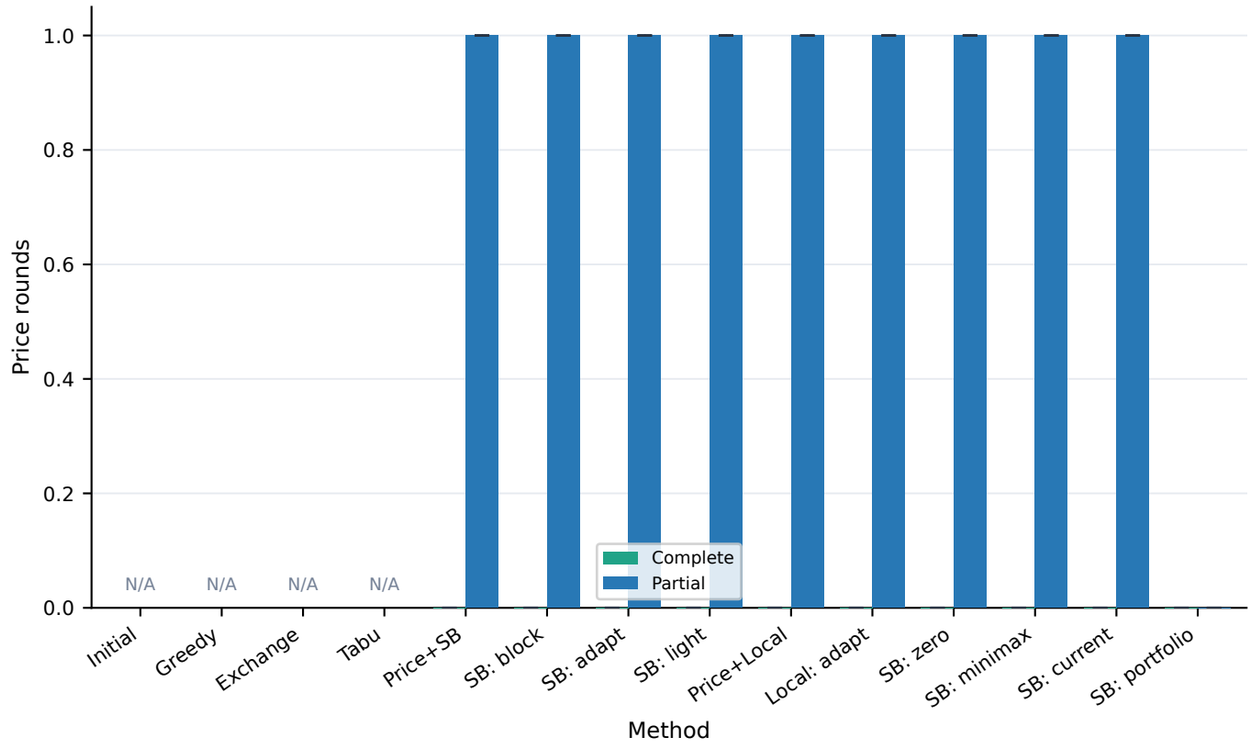


custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

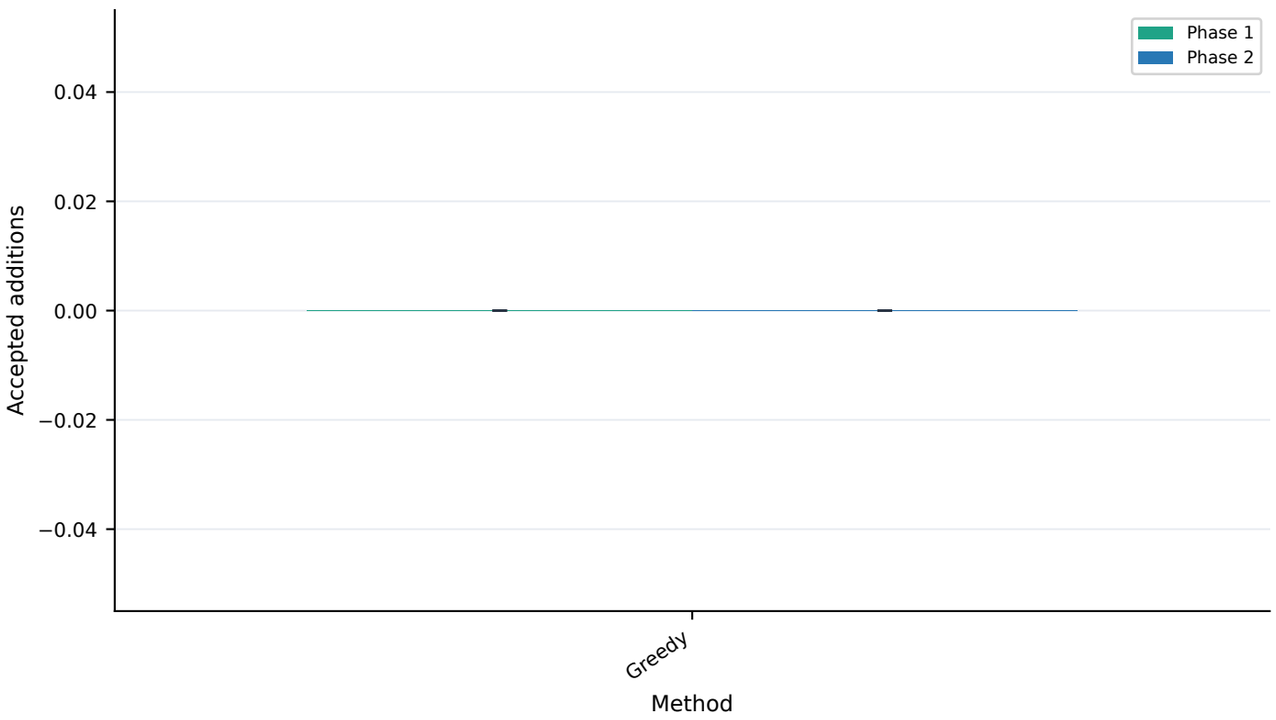
Price rounds attempted, including partial



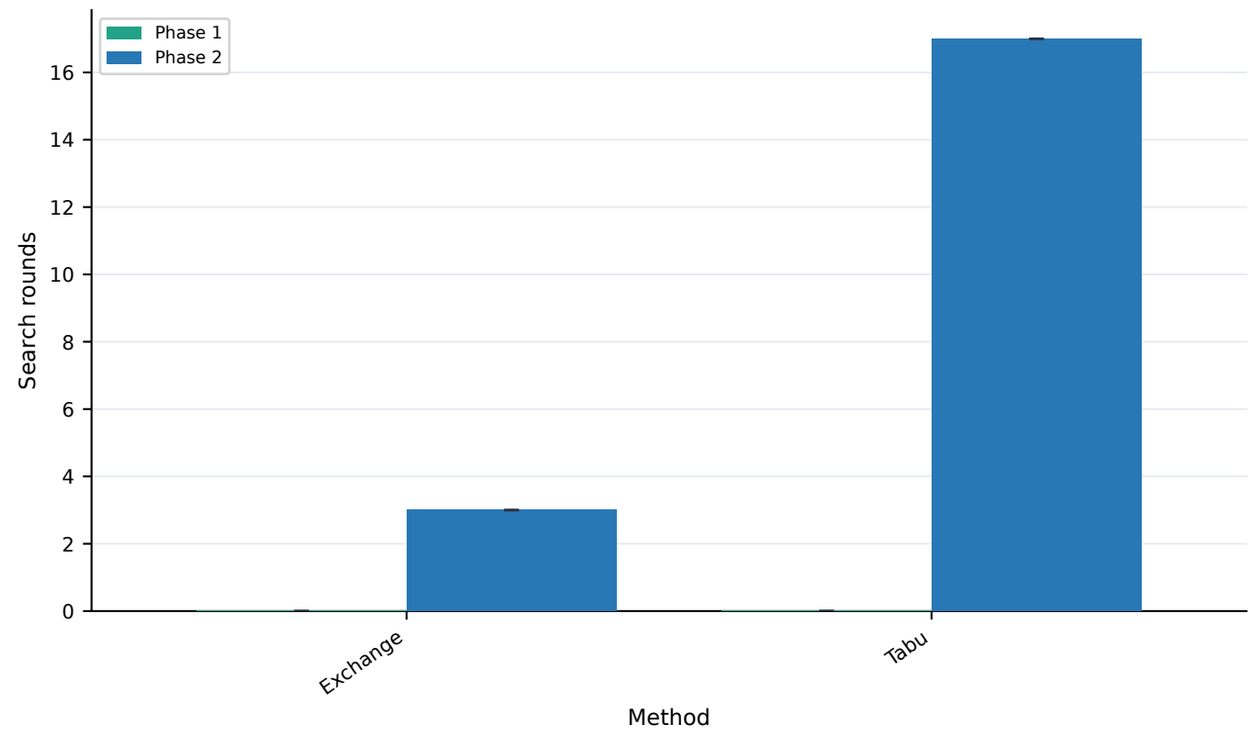
Complete versus partial price rounds



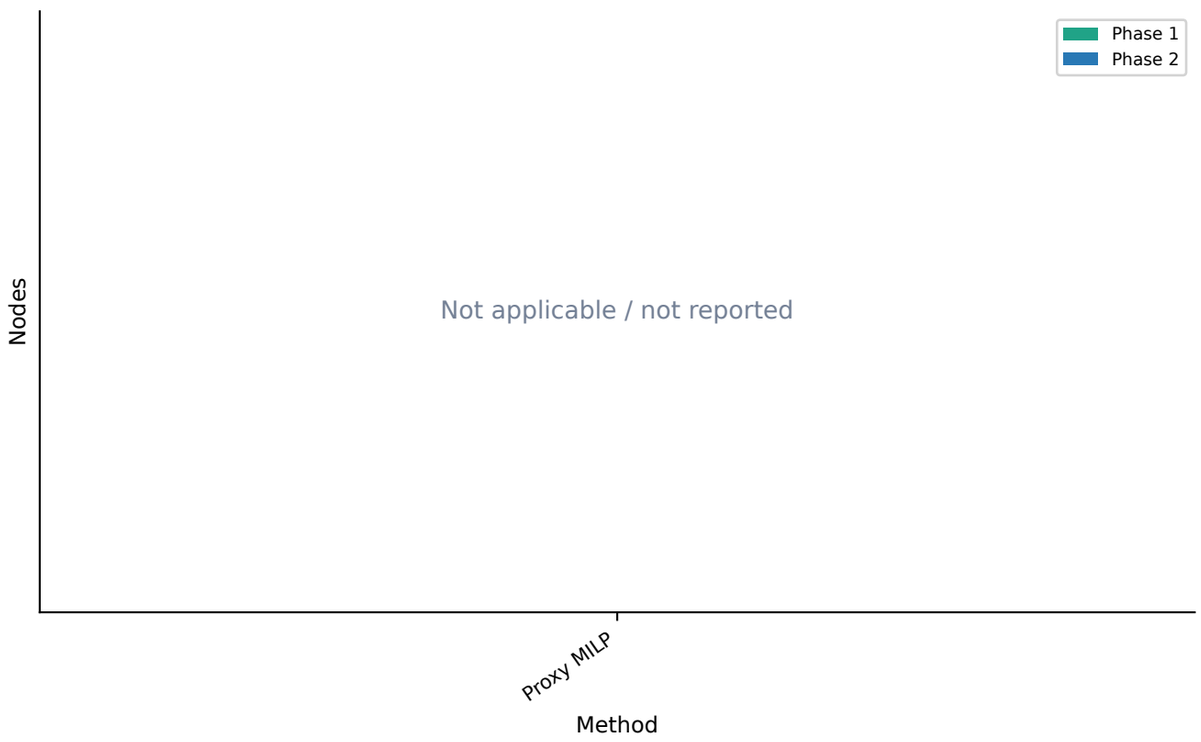
Greedy accepted additions



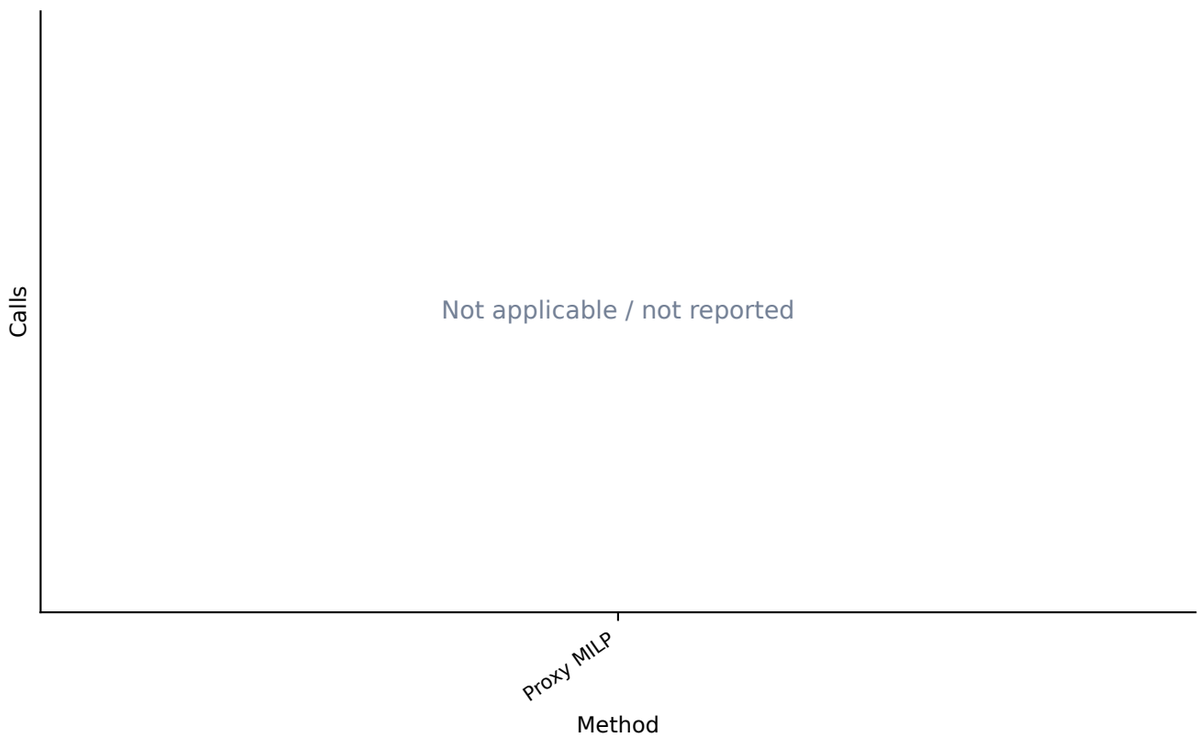
Original-objective neighborhood rounds



MILP branch-and-bound nodes



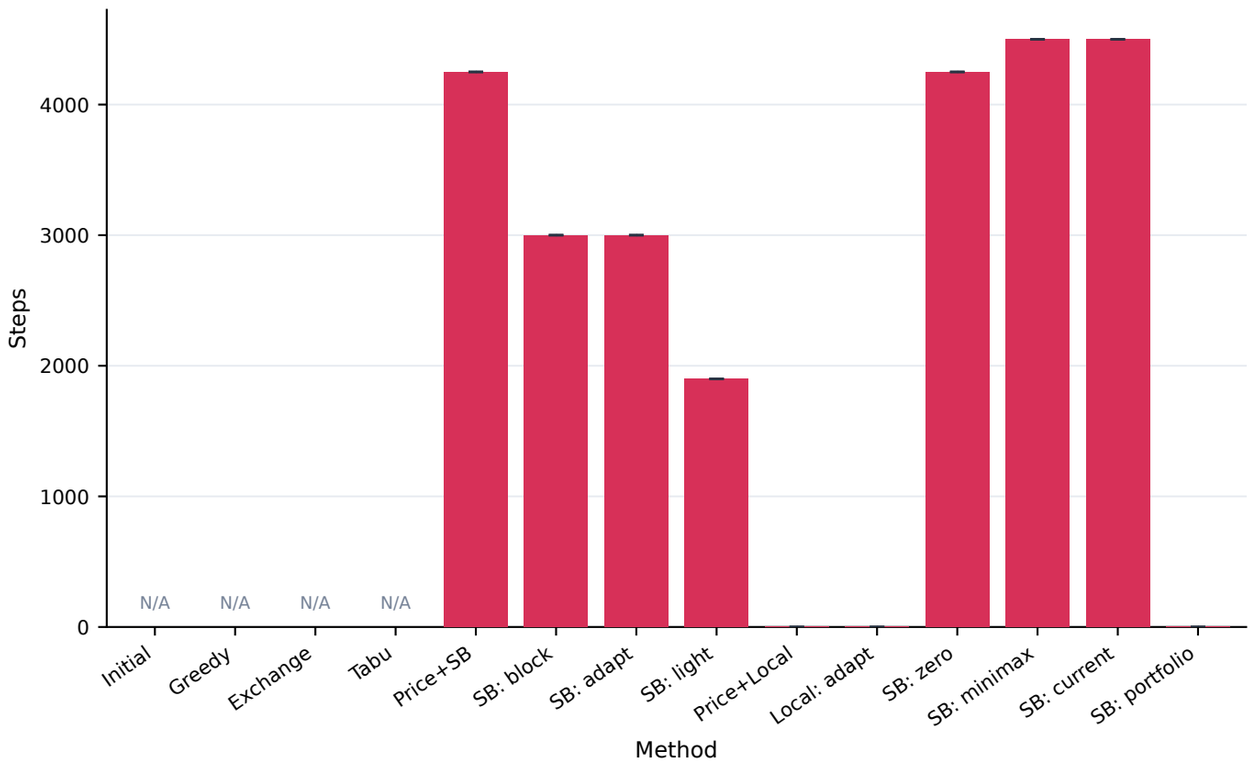
MILP solver invocations (separate unit)



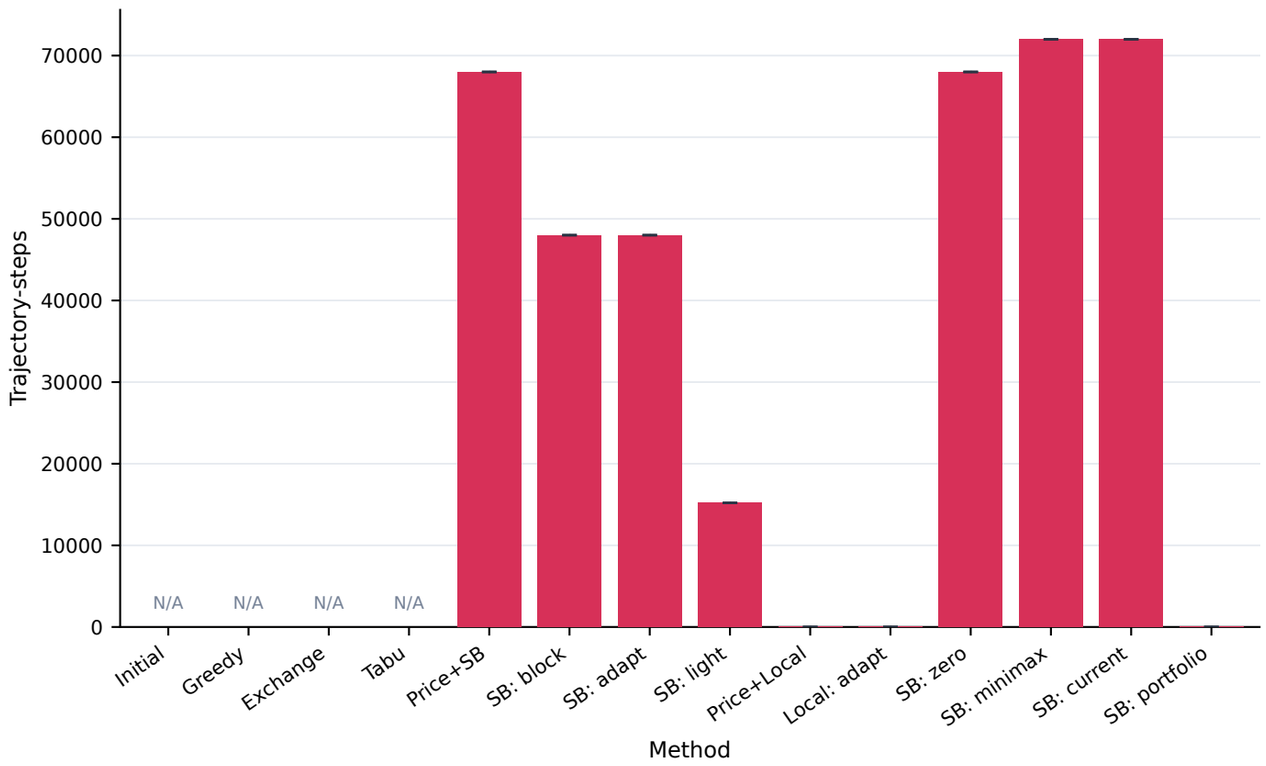
GLPK does not report node counts; missing values are not zero.

custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

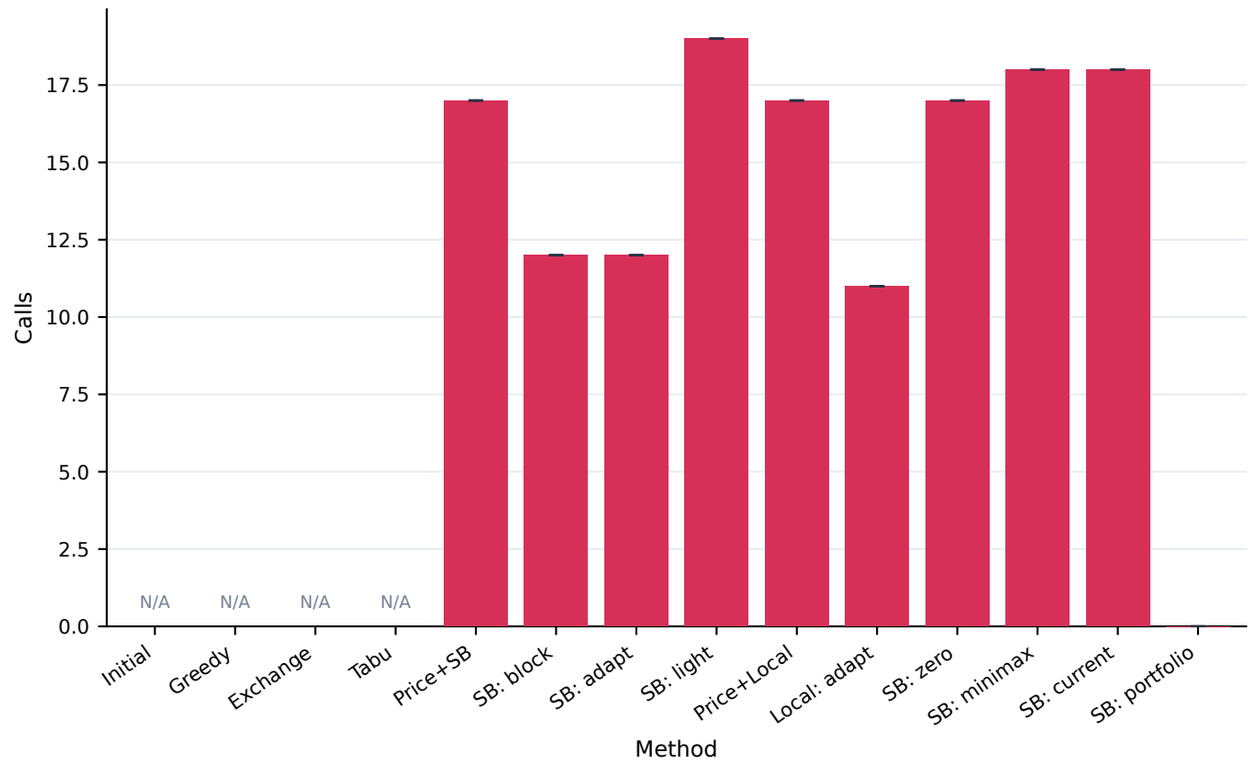
SB integration steps summed over all calls



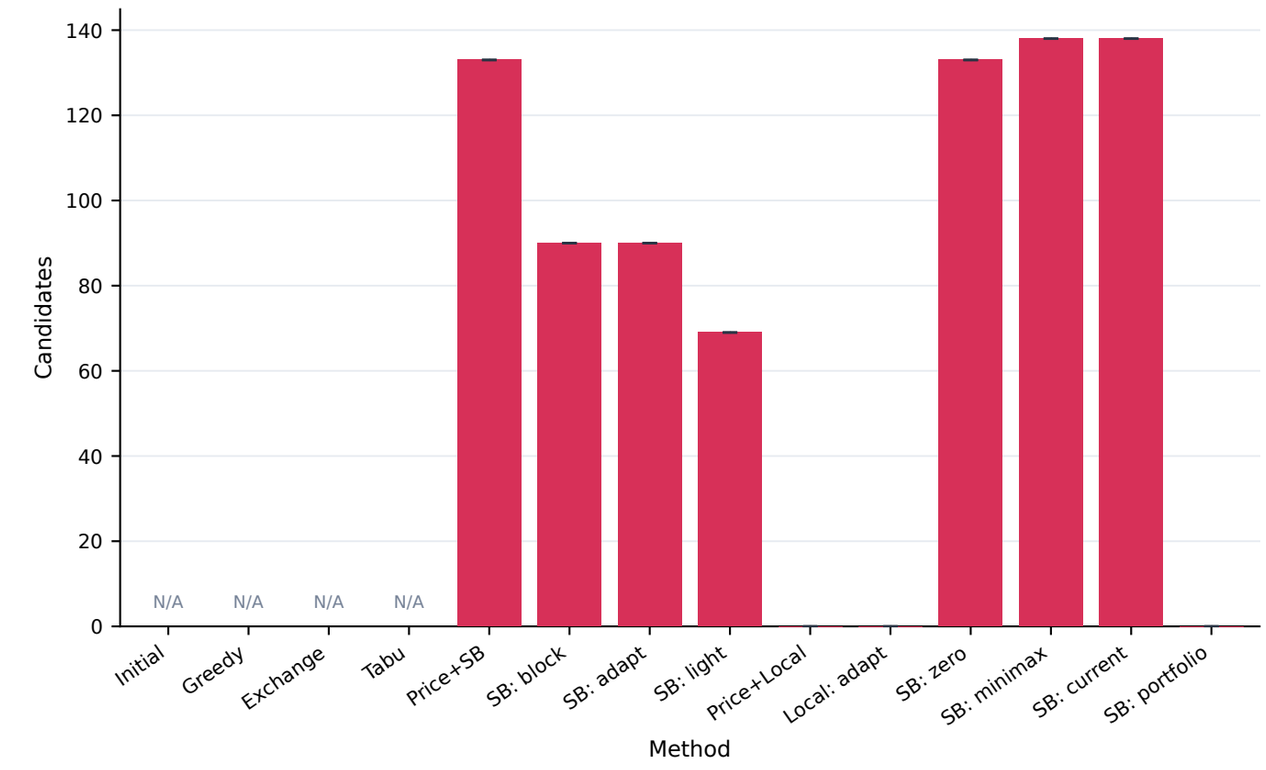
SB steps multiplied by trajectory count



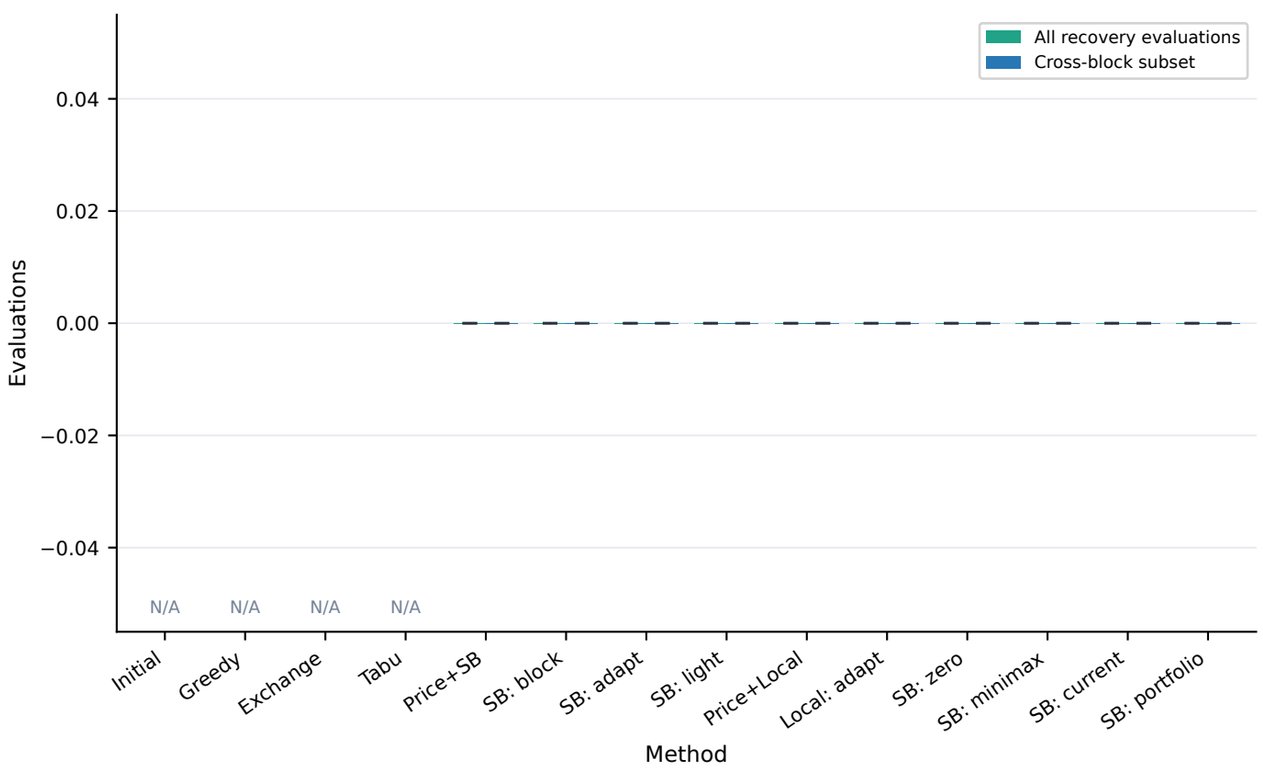
Resource subproblems solved



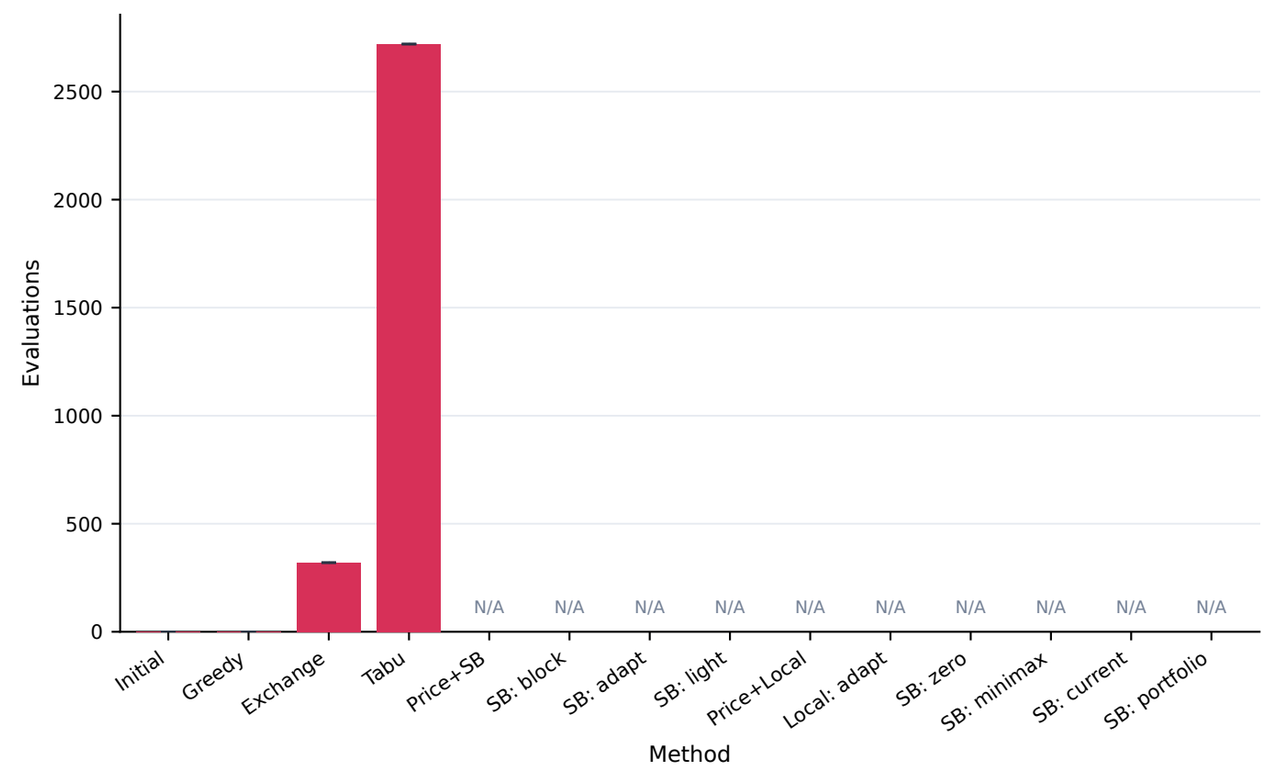
Candidates changed by hard repair



Recovery evaluations (cross is a subset)

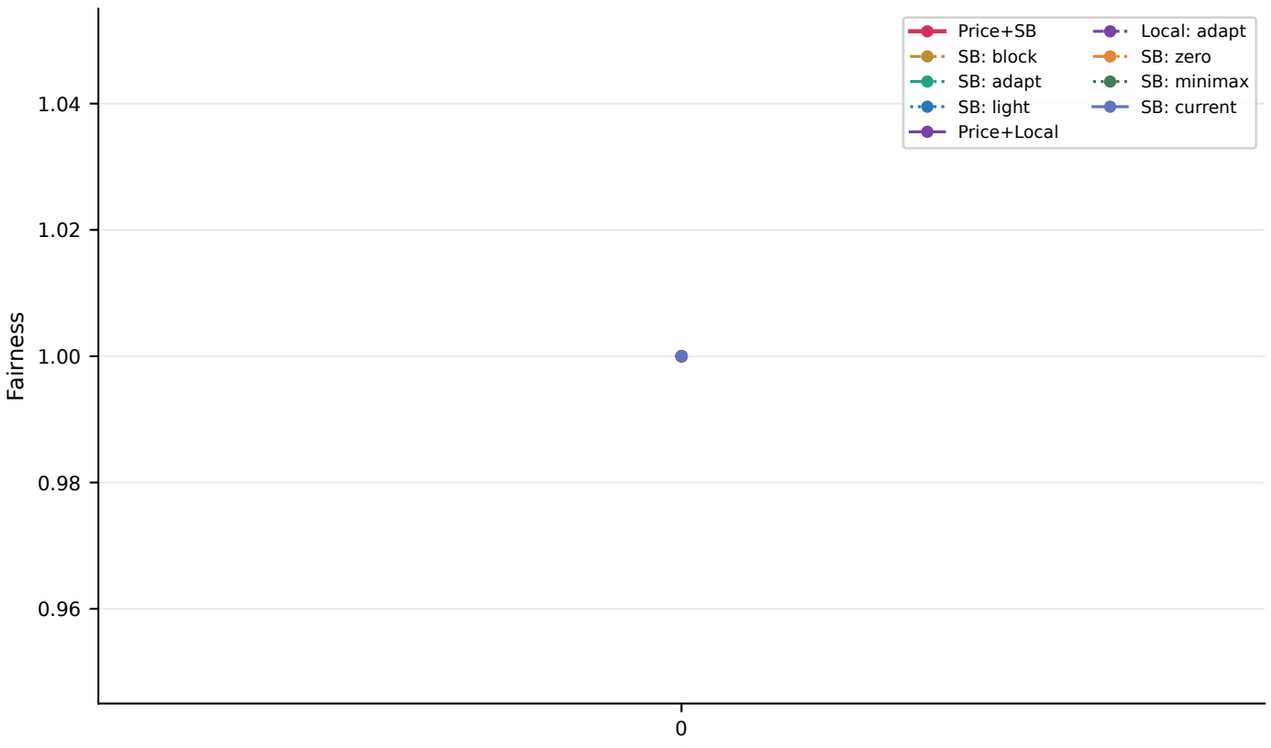


Original baseline search evaluations

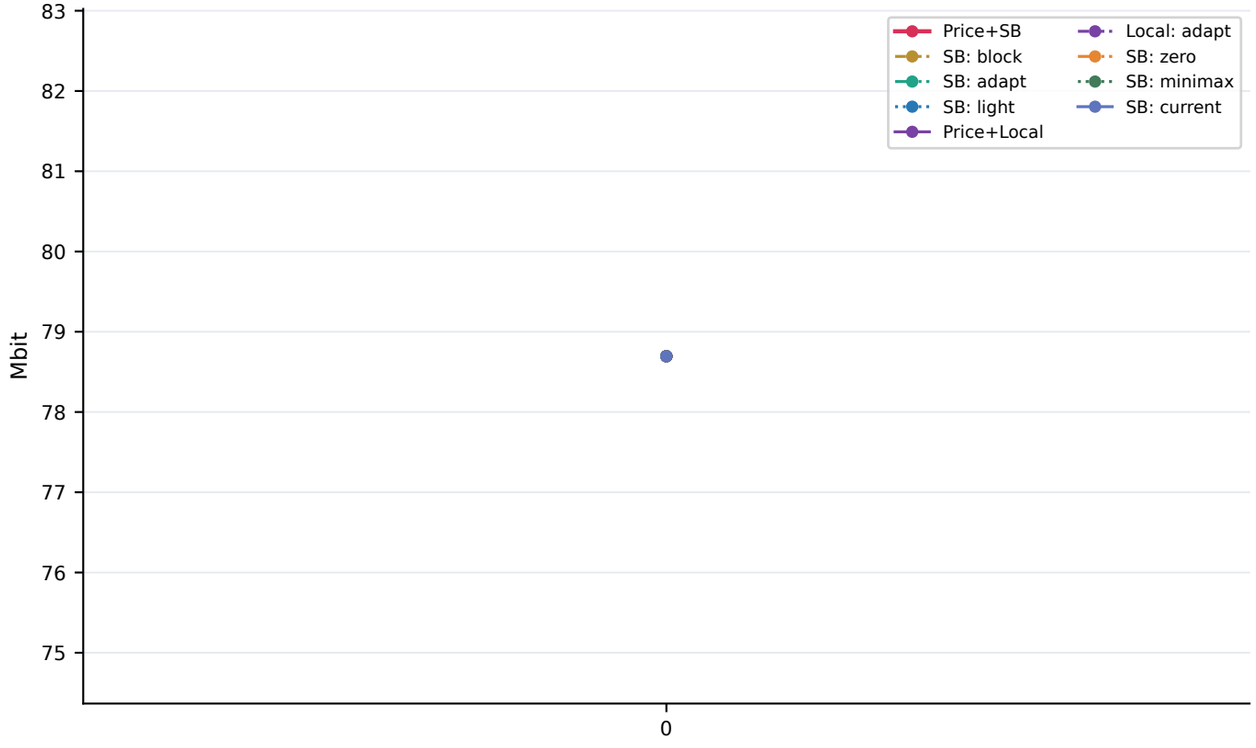


custom | first requested seed 71669974 | certified period service | price methods; portfolio branches excluded

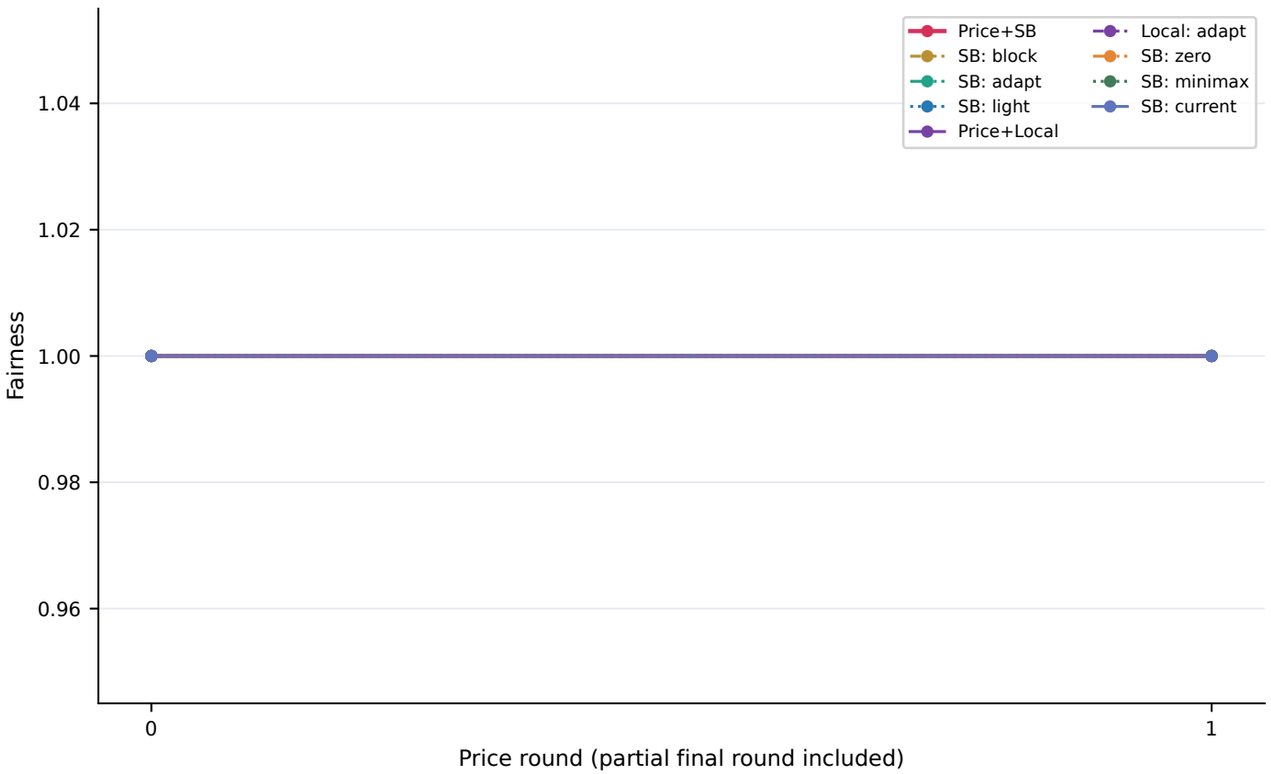
Saved original fairness: phase 1



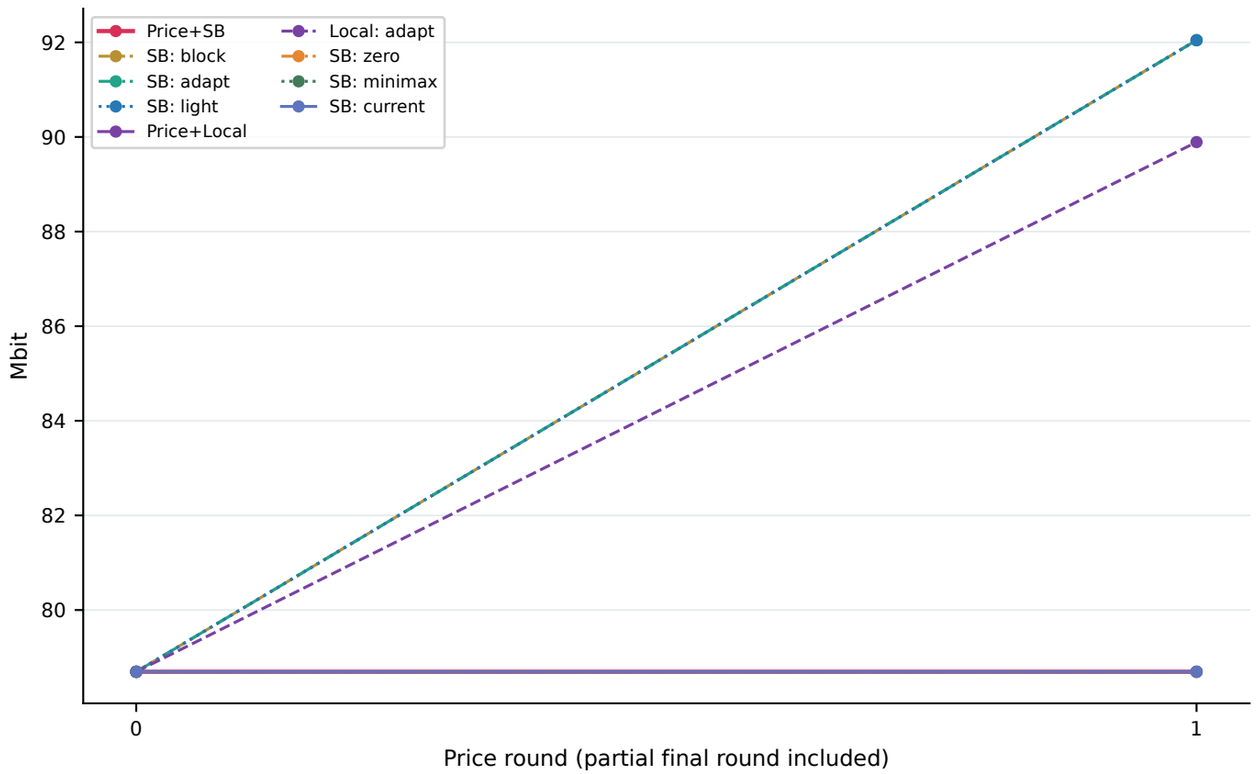
Diagnostic service: feasibility search (may decrease)



Saved original fairness: phase 2

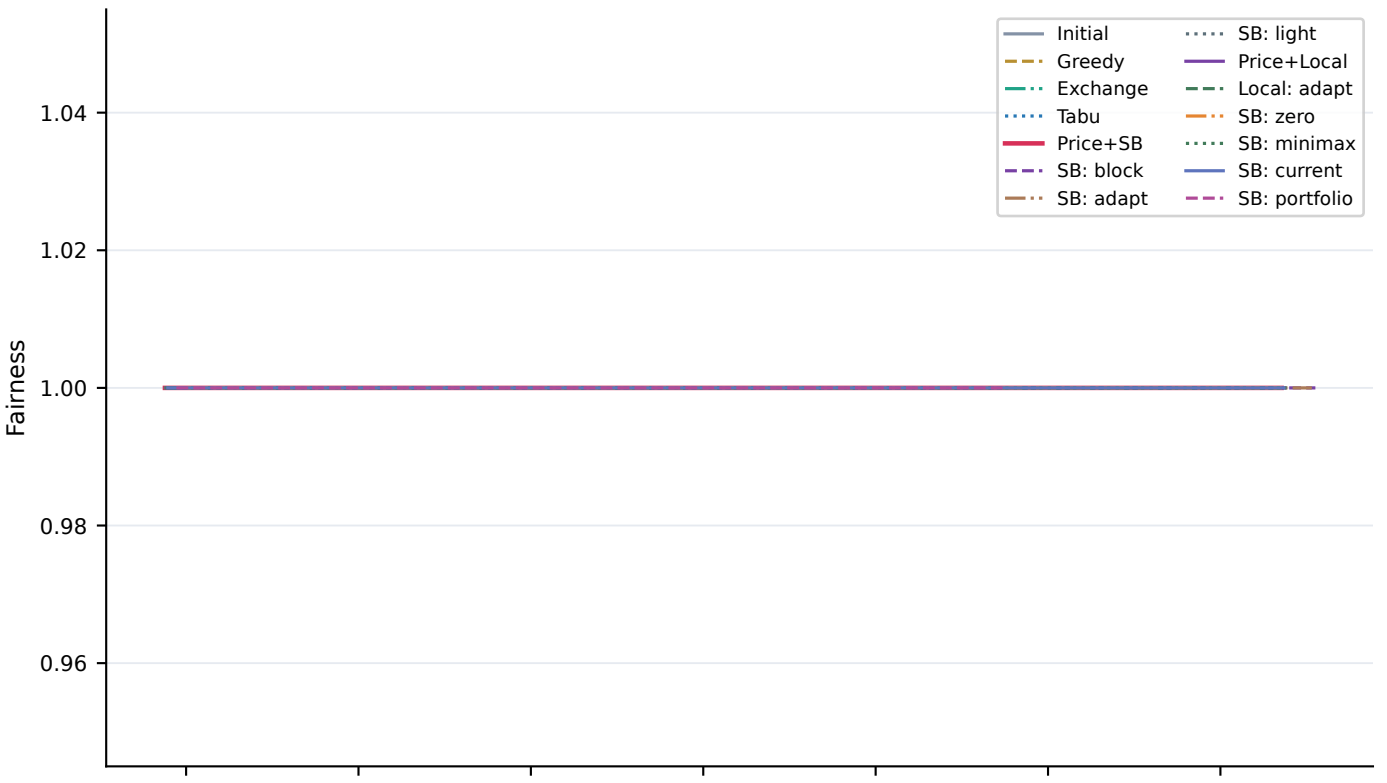


Saved throughput: phase 2

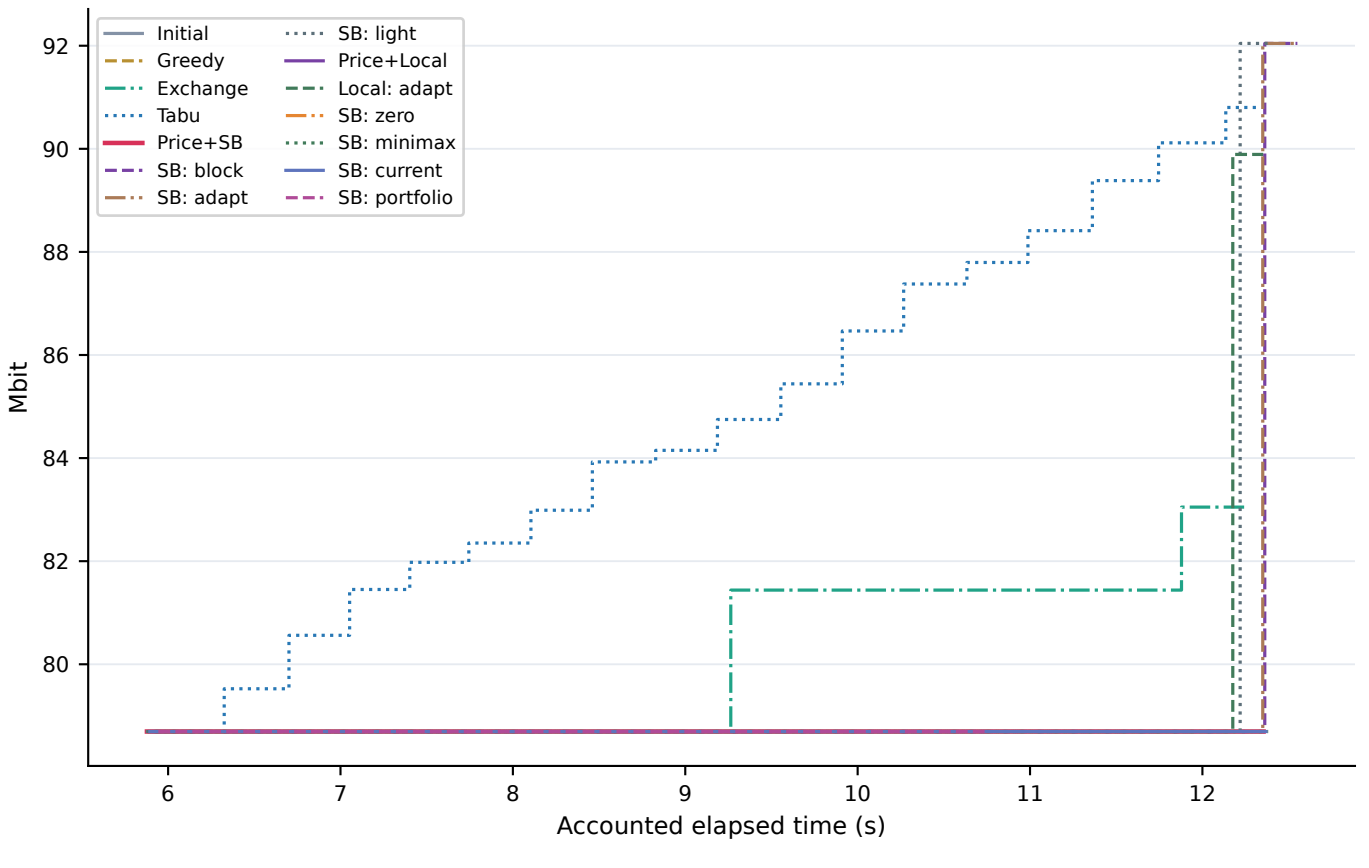


custom | first requested seed 71669974 | certified period service | recorded checkpoints; no solver-internal interpolation

Original fairness versus accounted time

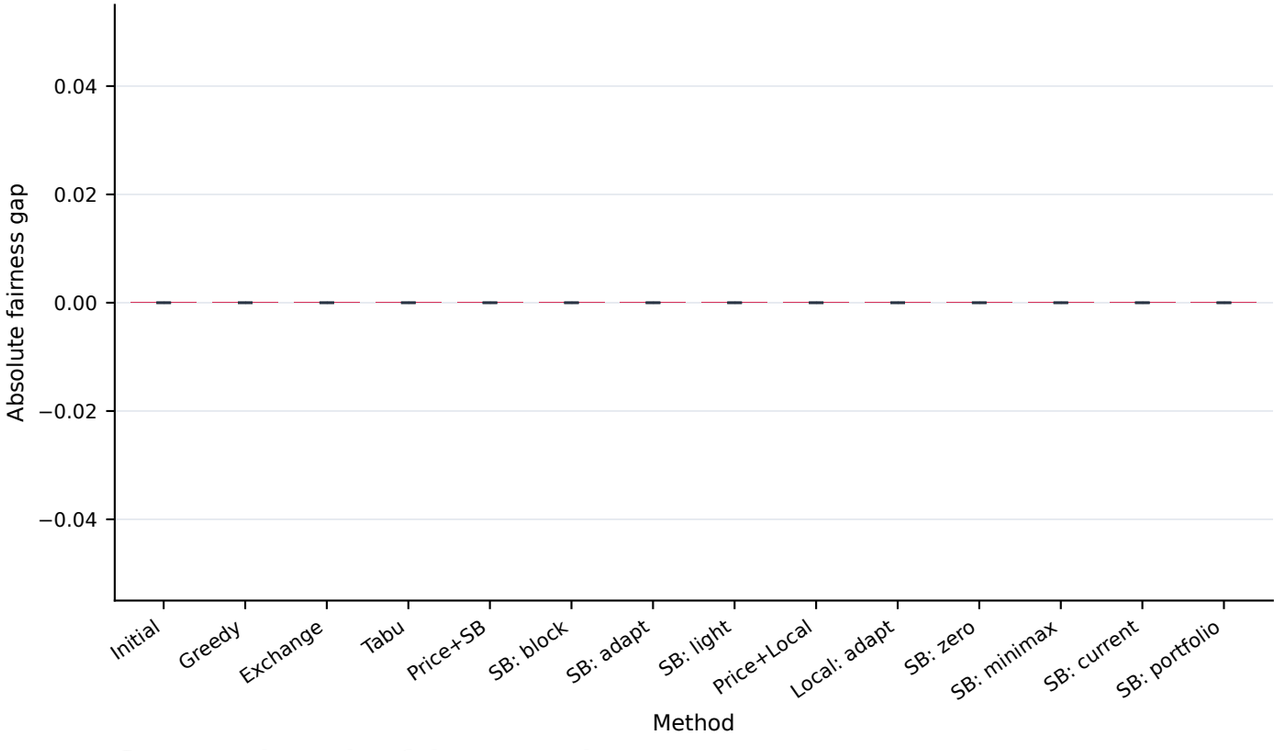


Diagnostic service versus accounted time (check full QoS)

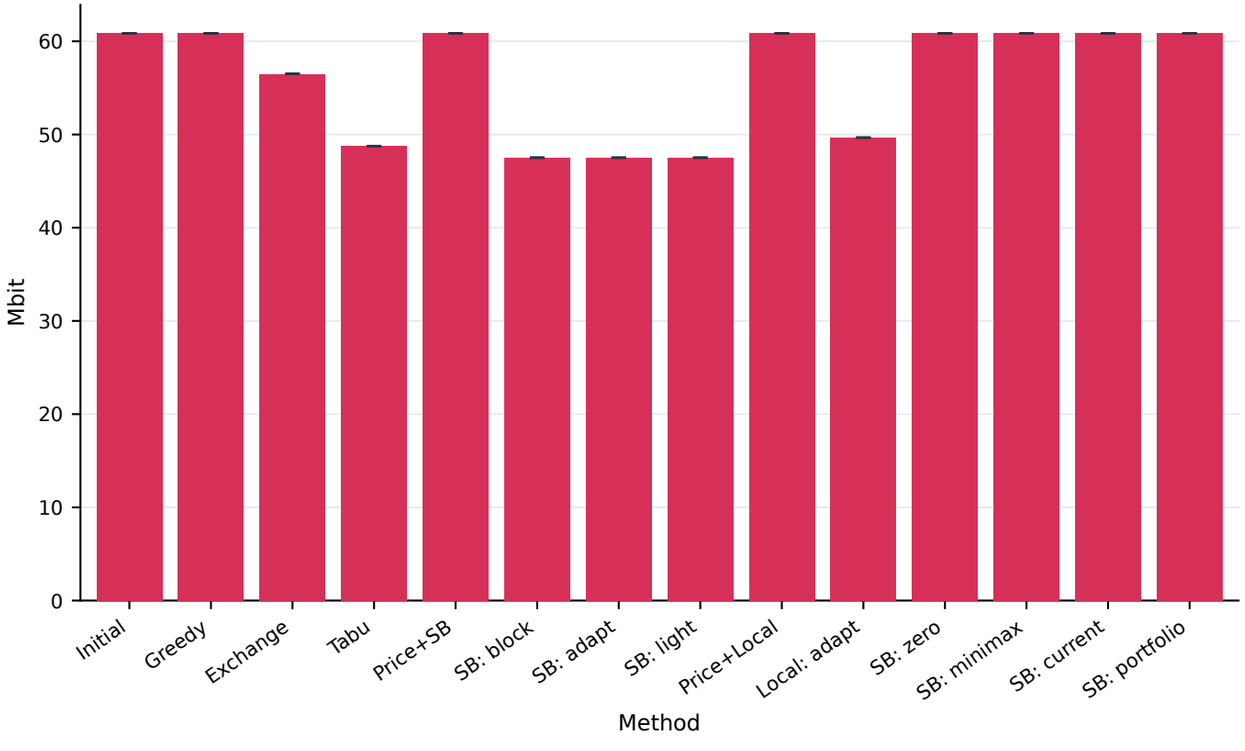


custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

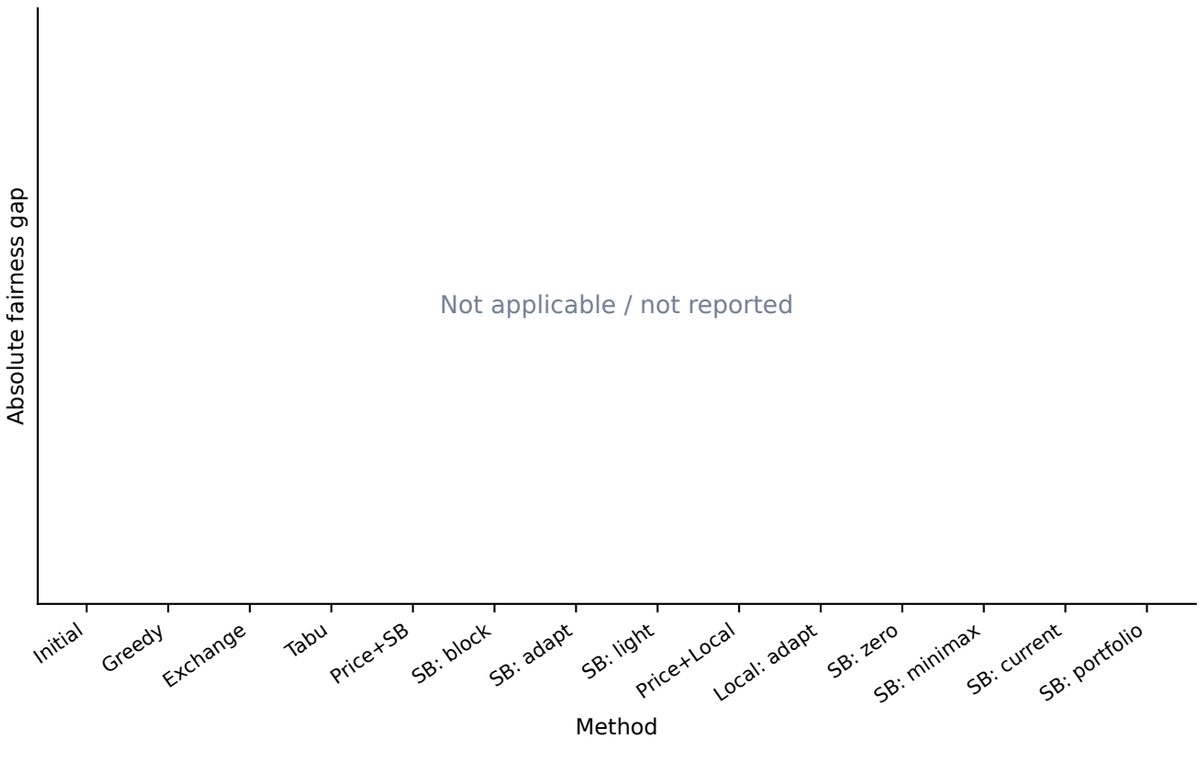
Valid original fairness bound minus achieved value



Valid throughput gap at the recorded required floor

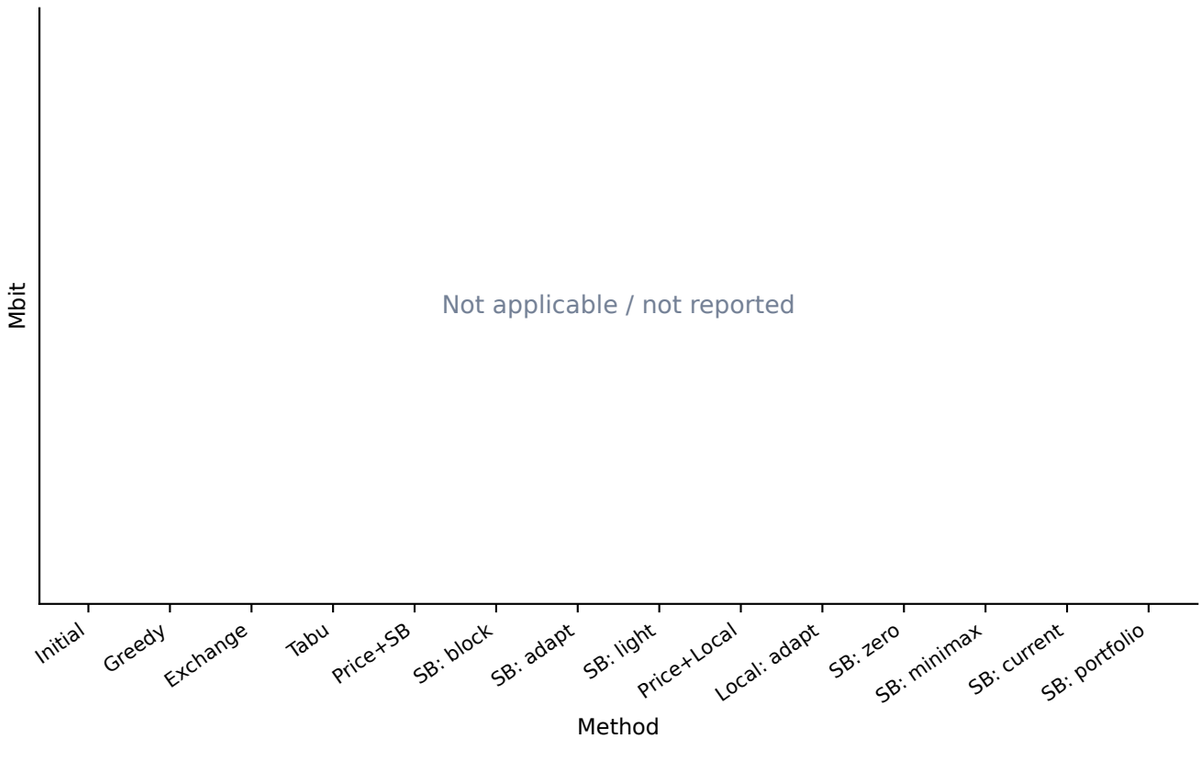


Gap to exhaustive fairness optimum



Only a completed exhaustive search supplies an exact reference.

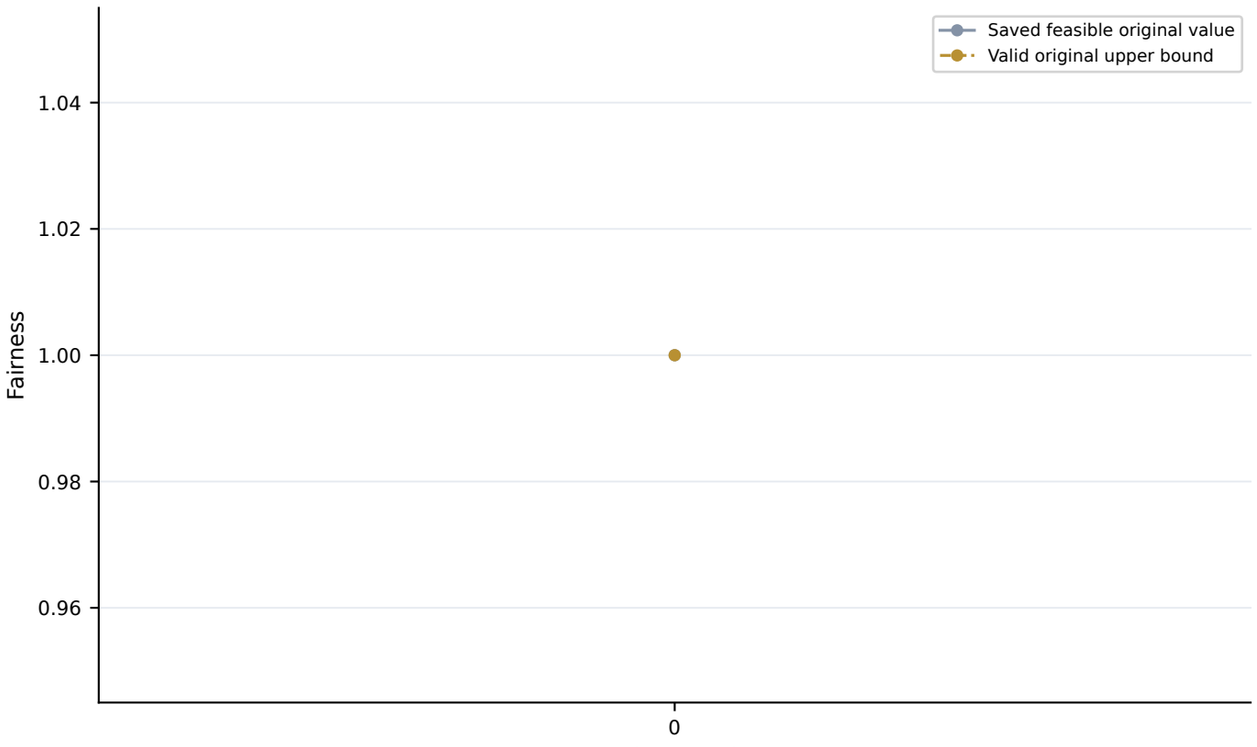
Exact throughput gap ONLY at exact fairness floor



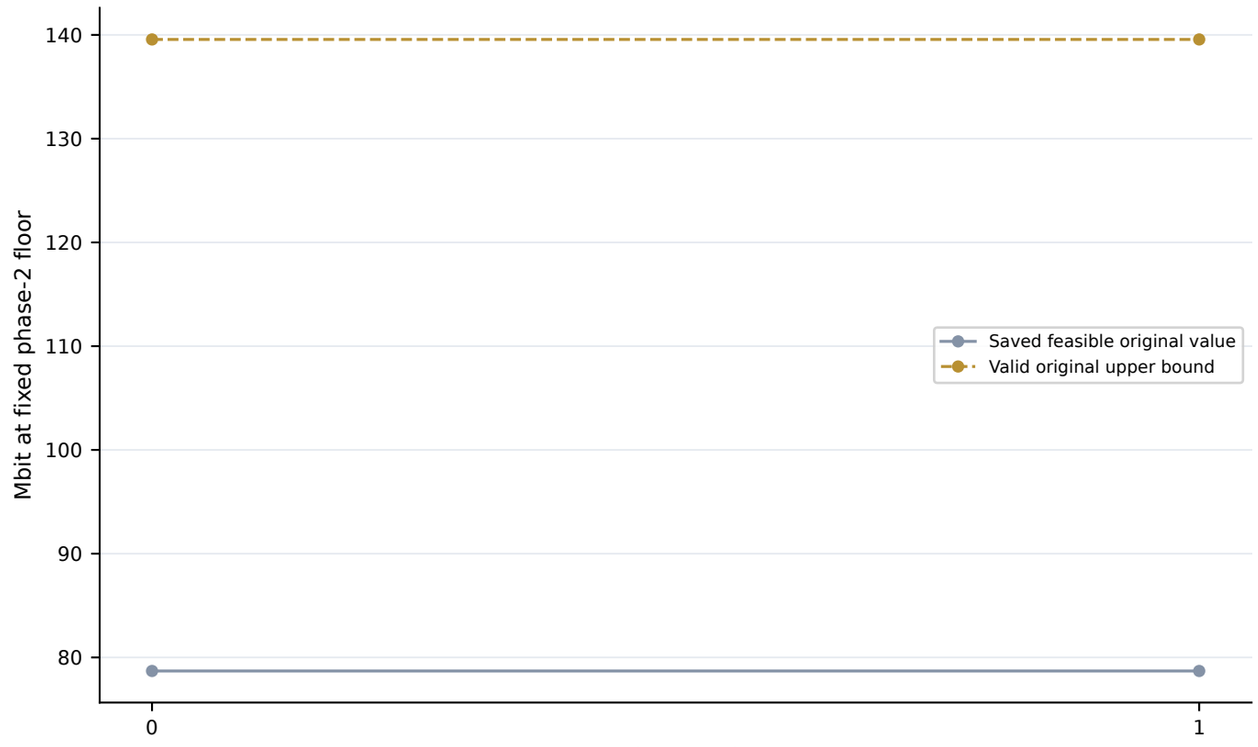
Different fairness floors are not comparable; these entries remain missing.

custom | first requested seed 71669974 | certified period service

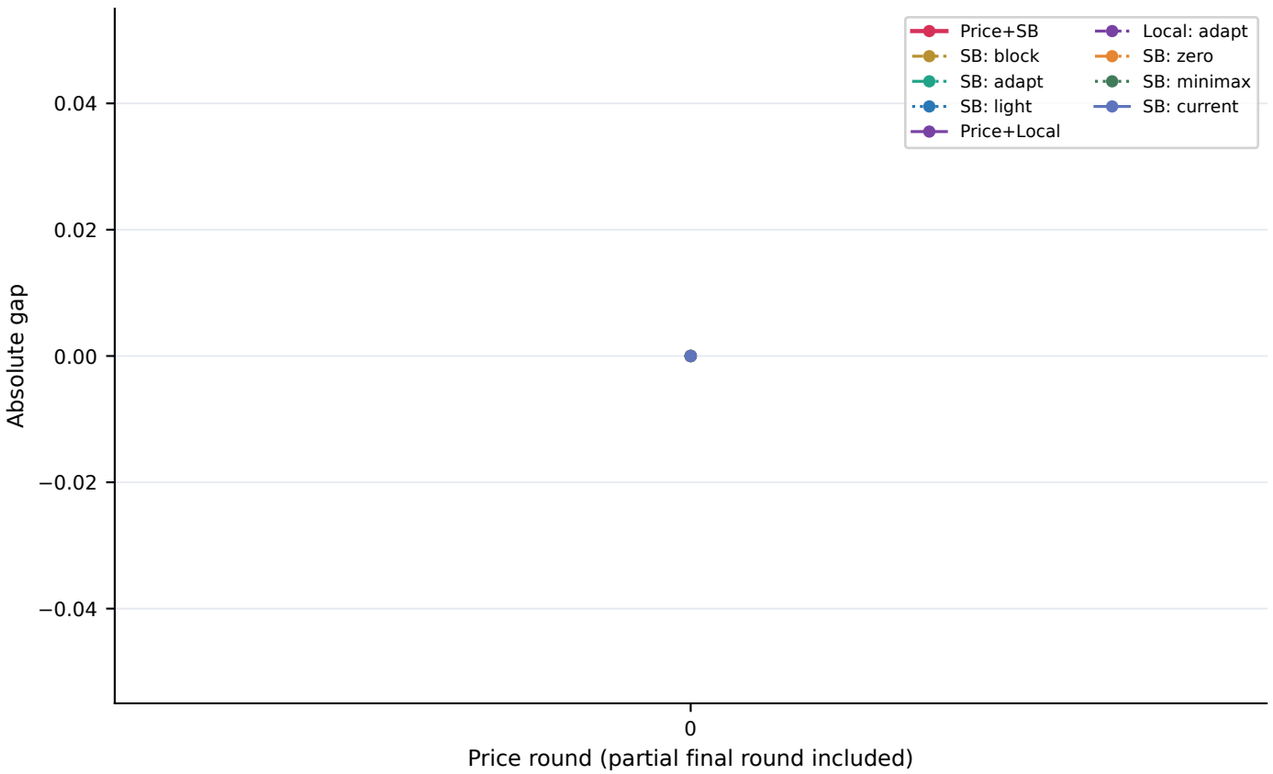
Price+SB valid original bounds: phase 1



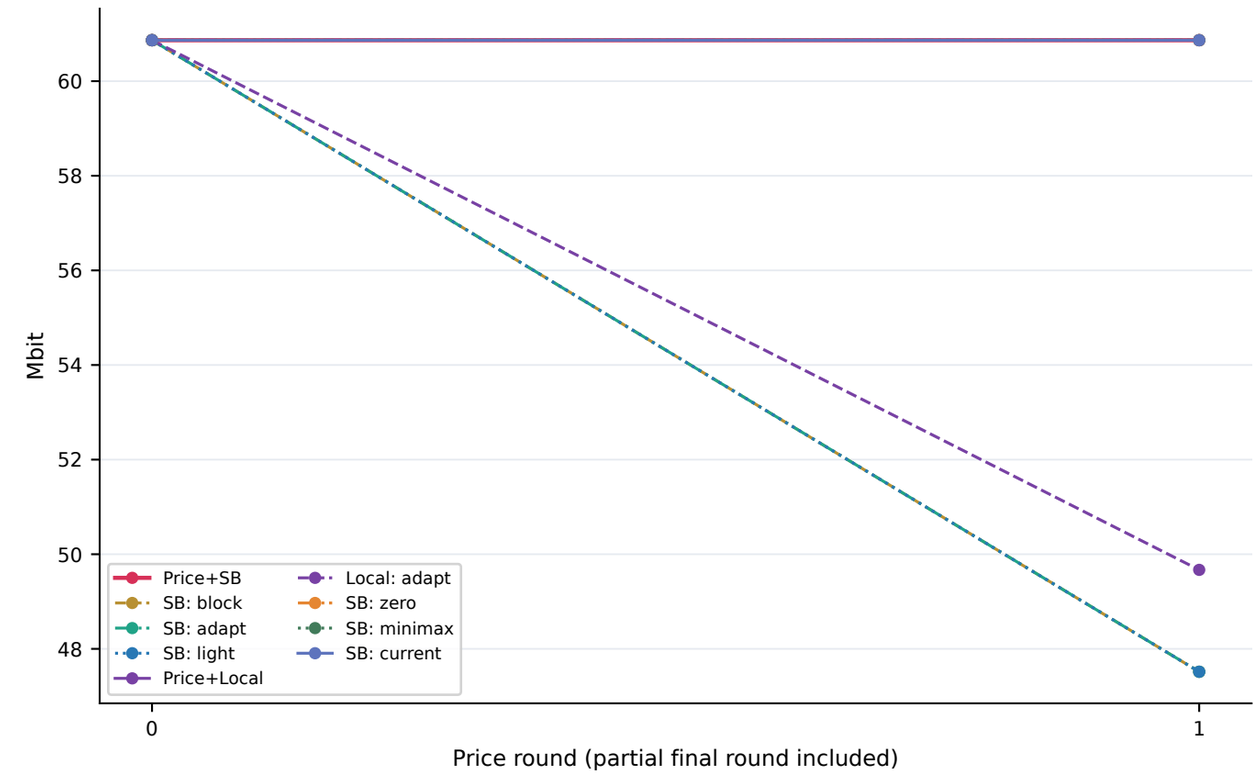
Price+SB valid original bounds: phase 2



Valid fairness gap during price search

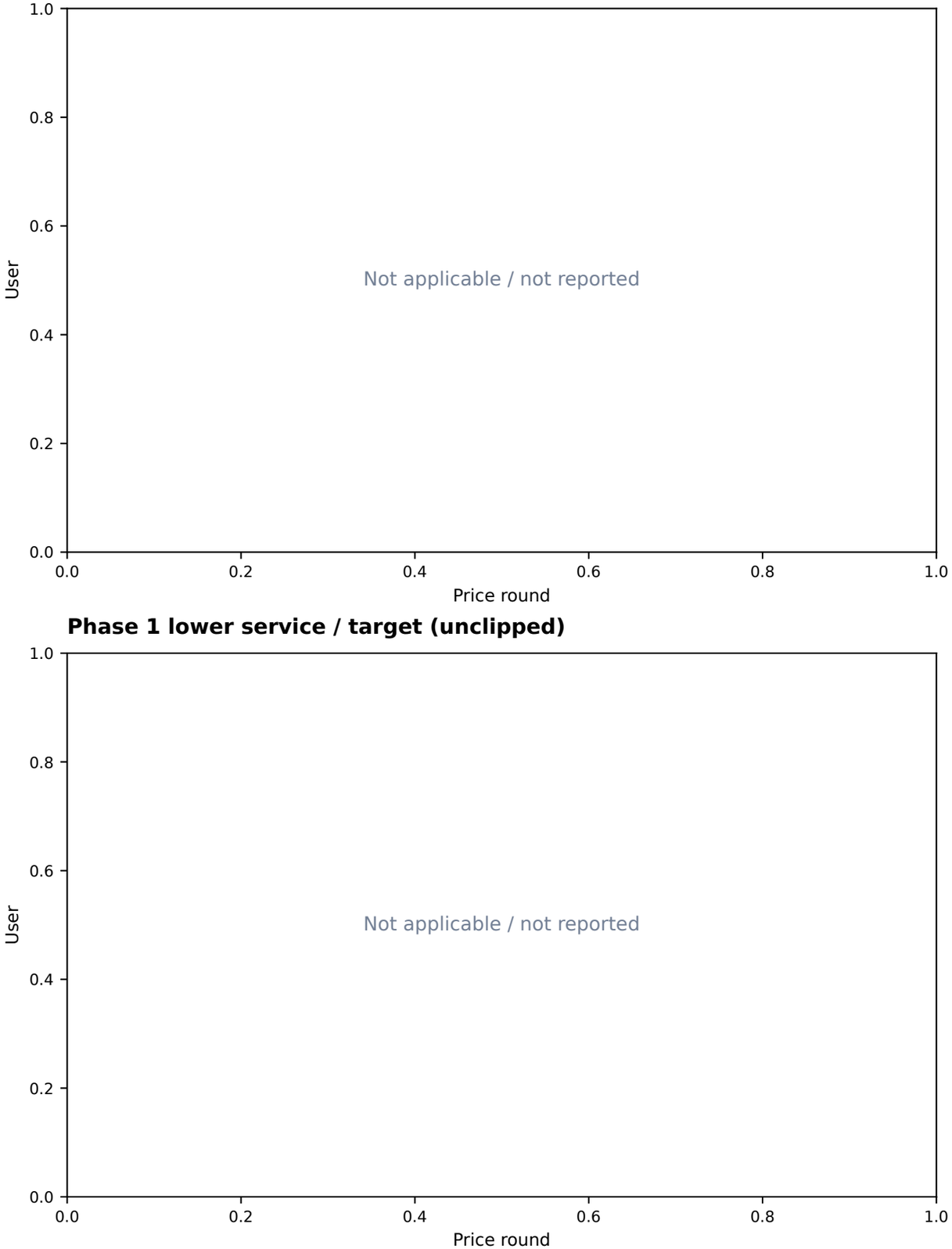


Valid throughput gap during price search

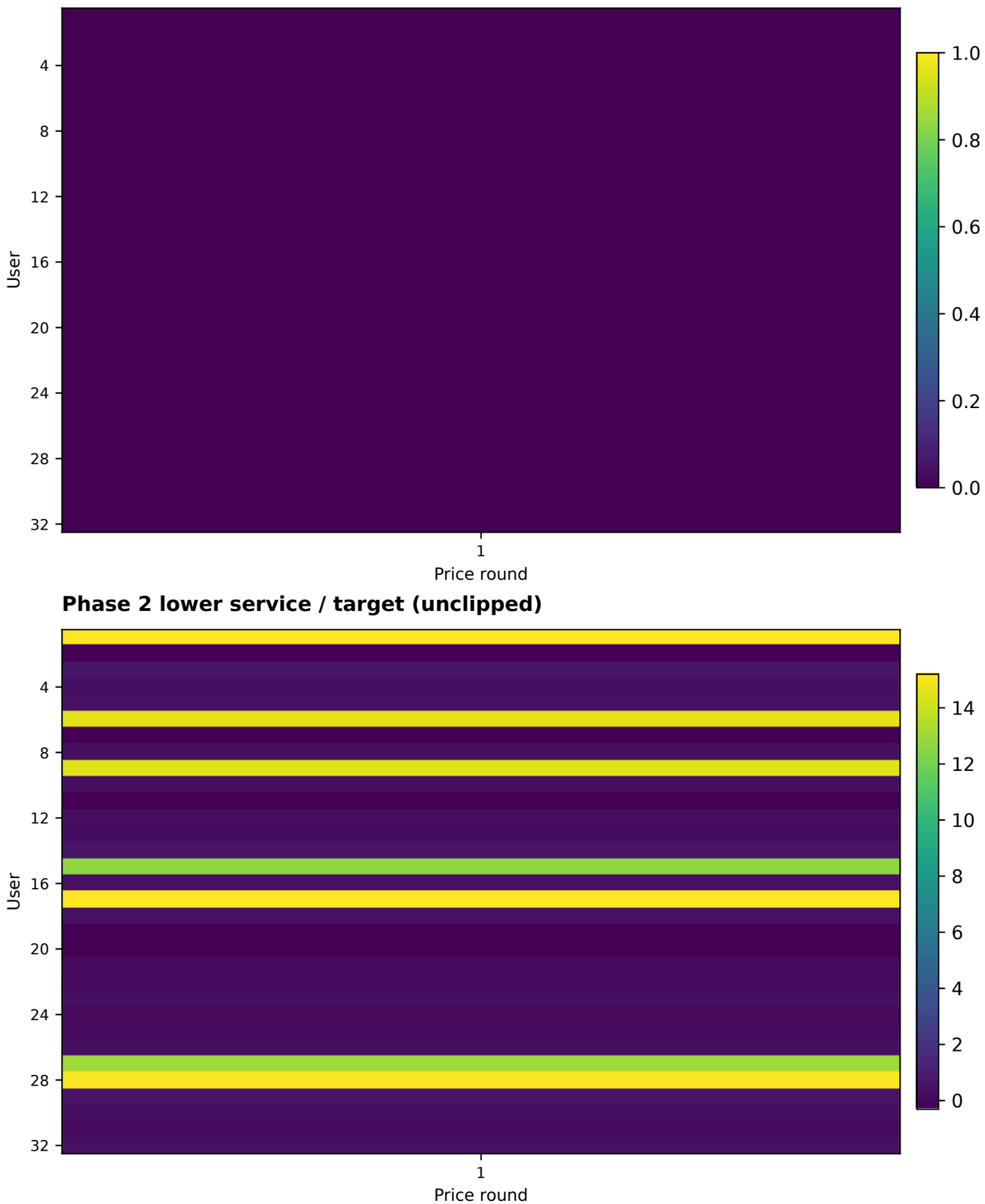


custom | first requested seed 71669974 | certified period service | Price+SB

Phase 1 nonnegative user prices



Phase 2 nonnegative user prices

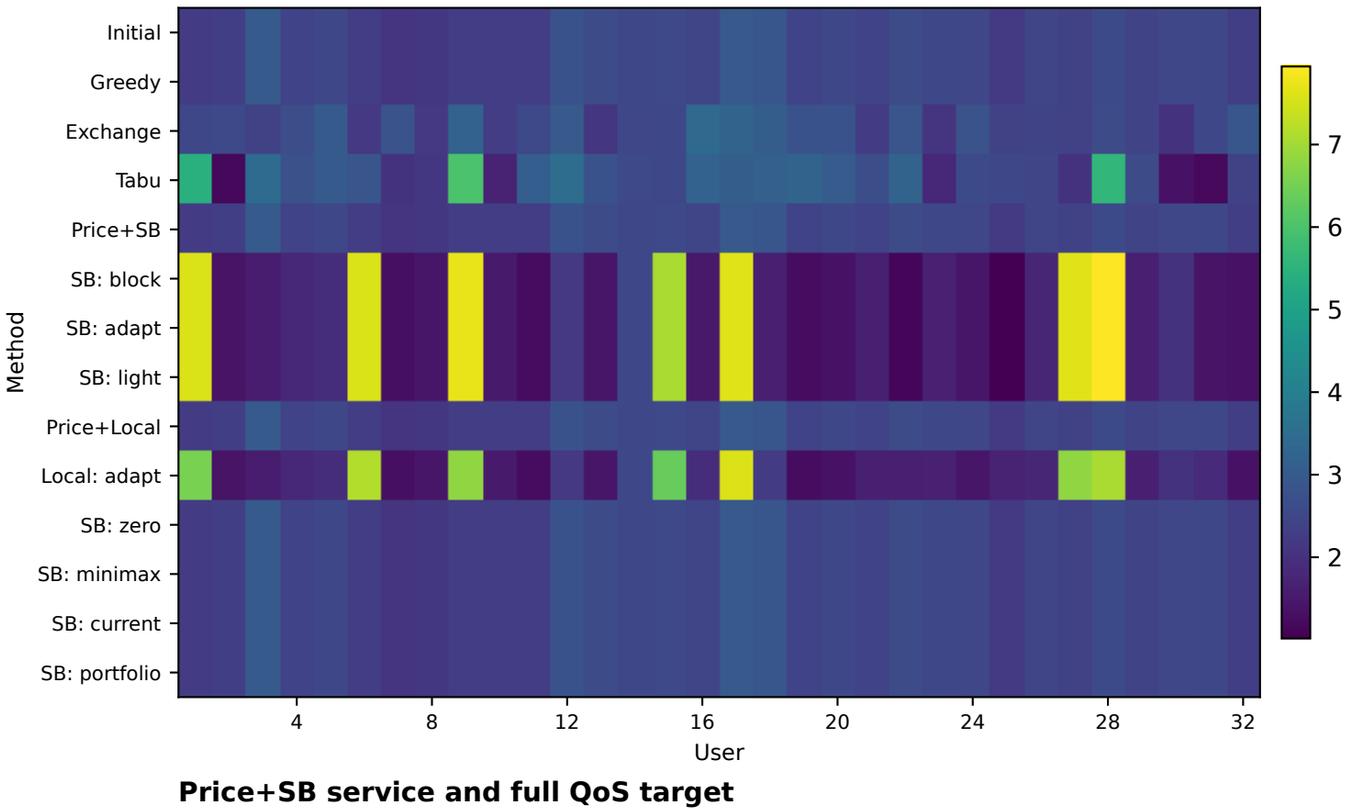


Phase 1 lower service / target (unclipped)

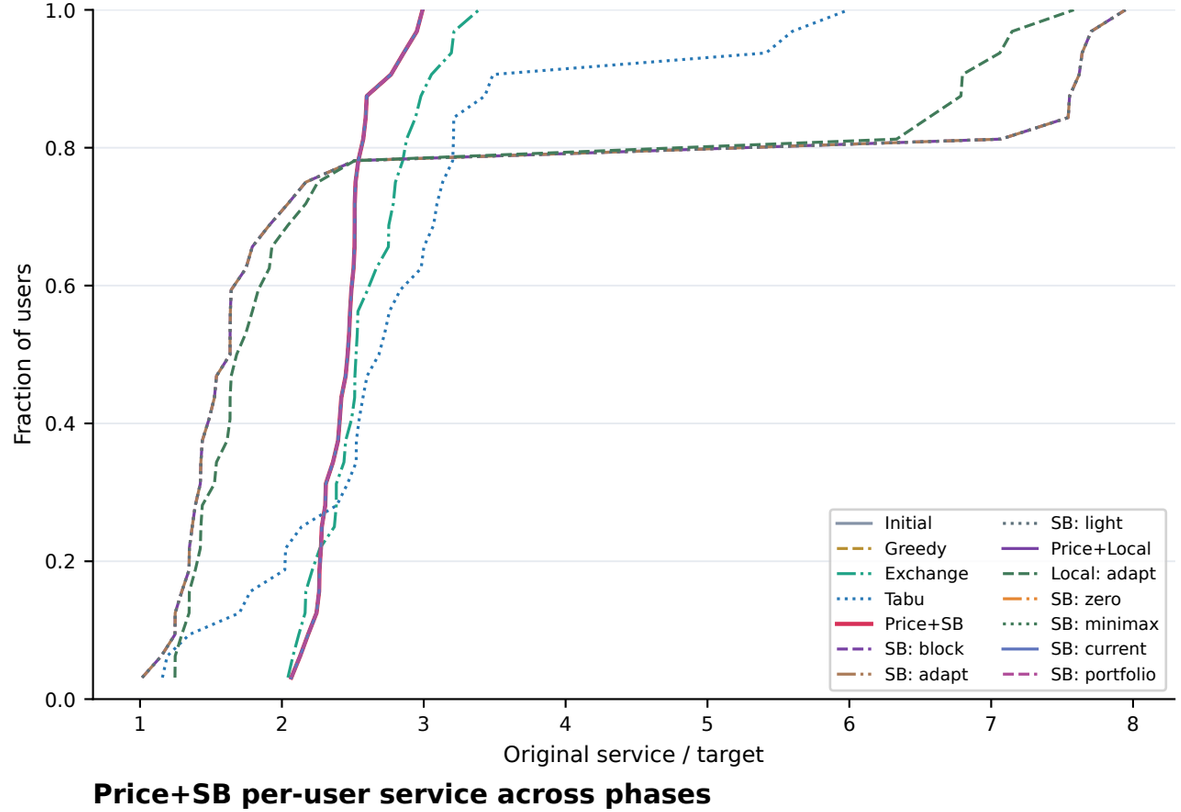
Phase 2 lower service / target (unclipped)

custom | first requested seed 71669974 | certified period service

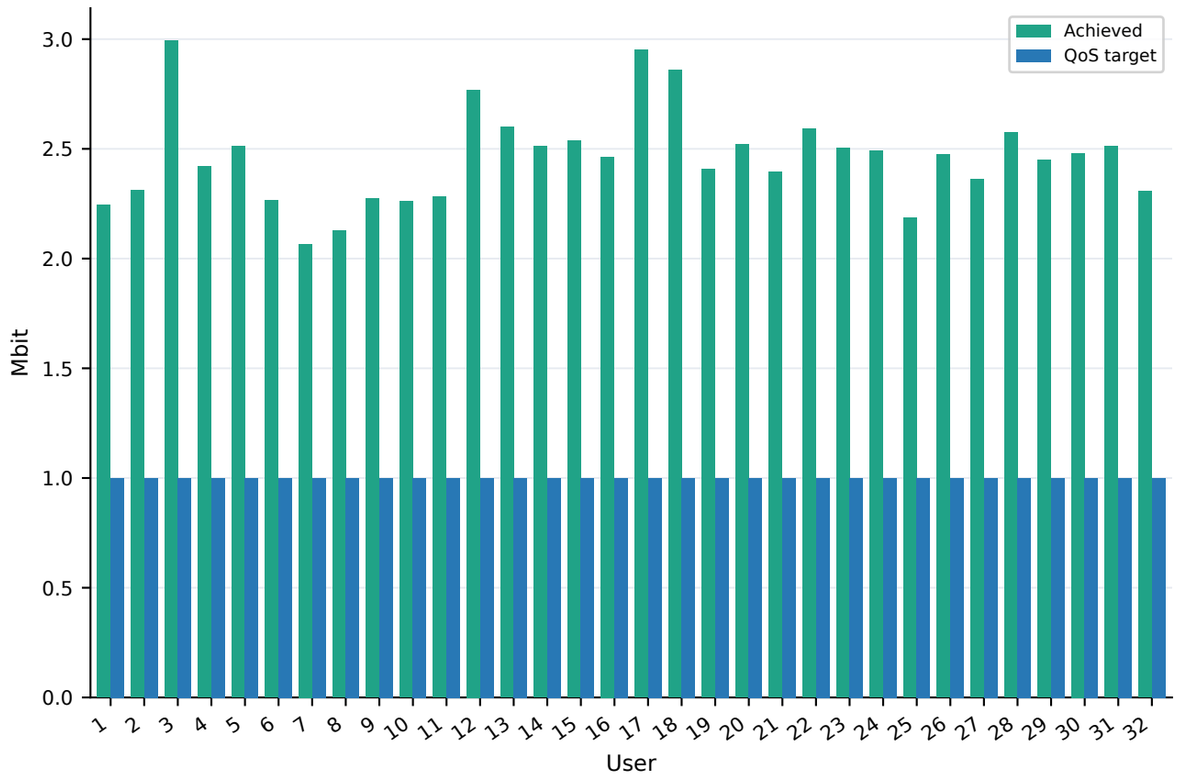
Every user original demand ratio



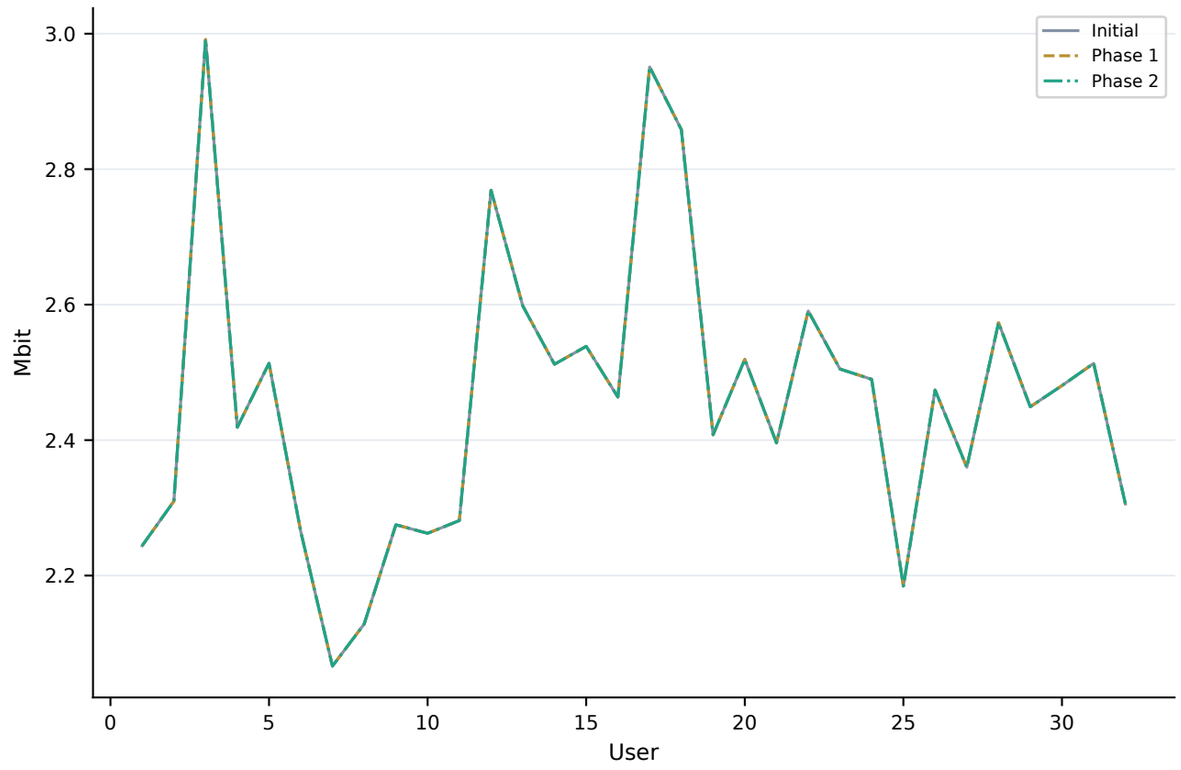
Empirical CDF of user demand ratios



Price+SB service and full QoS target

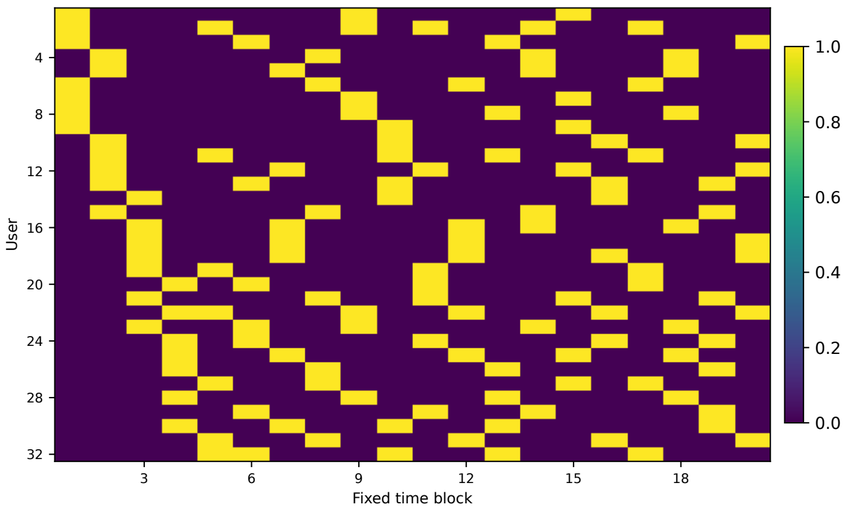


Price+SB per-user service across phases

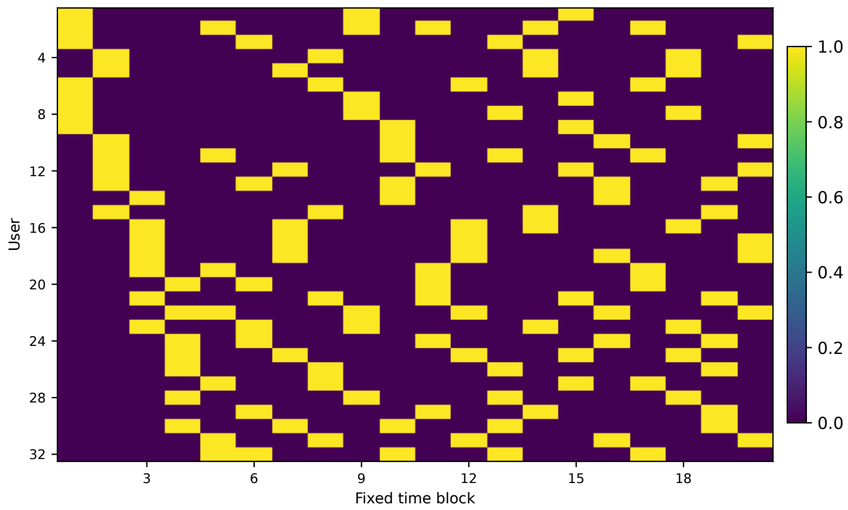


custom | first requested seed 71669974 | certified period service | 1=selected, 0=idle; one decision per fixed block

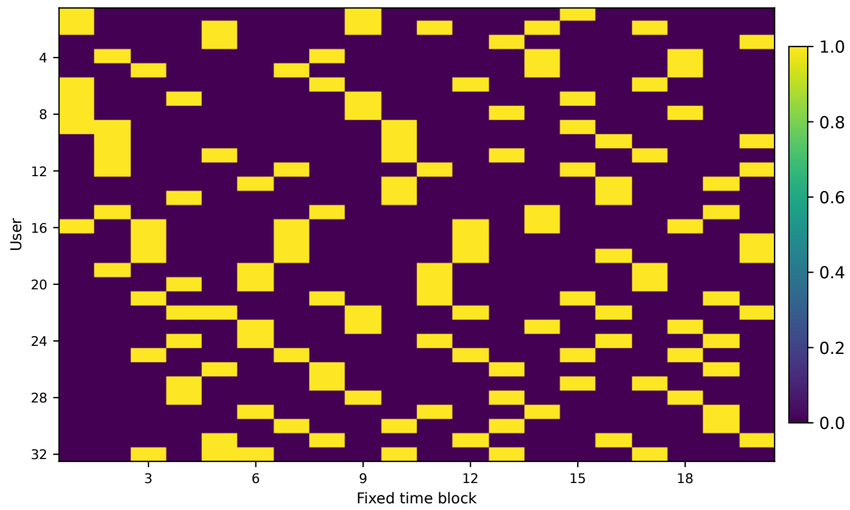
Initial



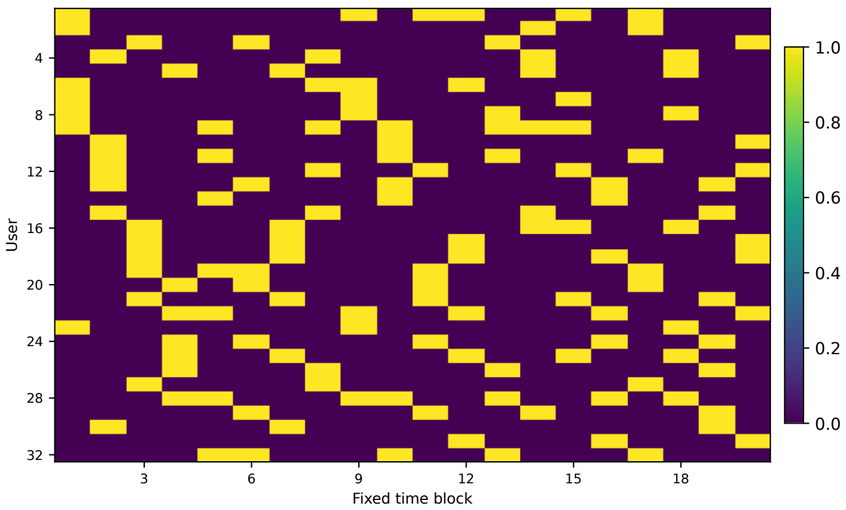
Greedy



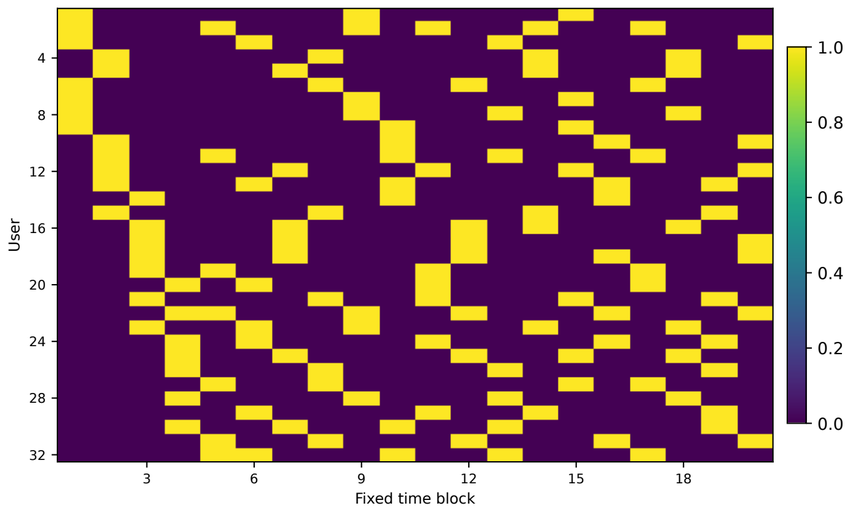
Exchange



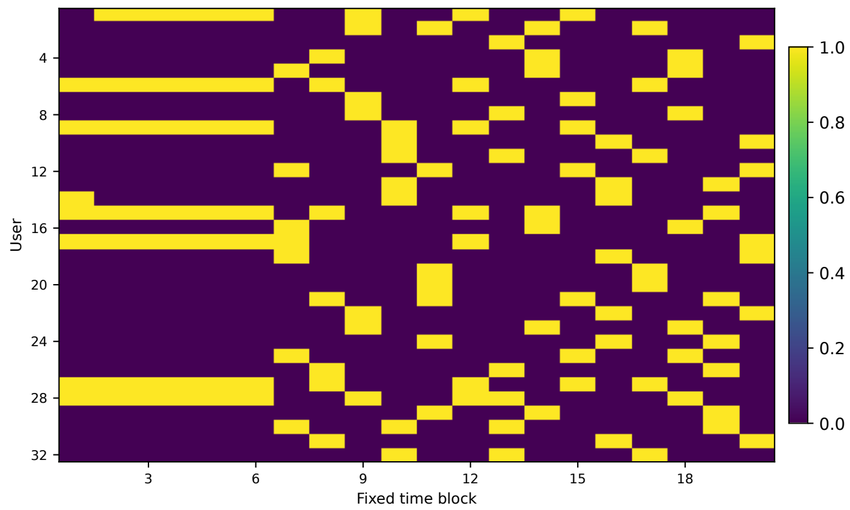
Tabu



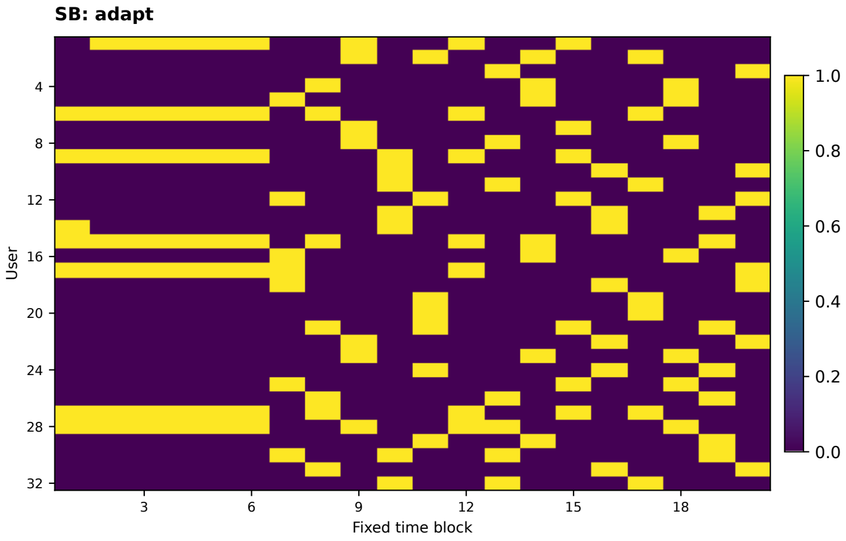
Price+SB



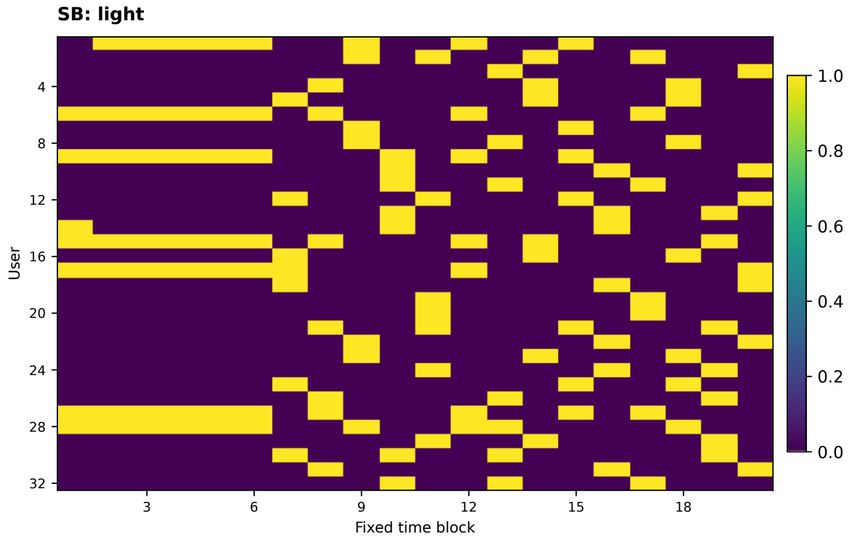
SB: block



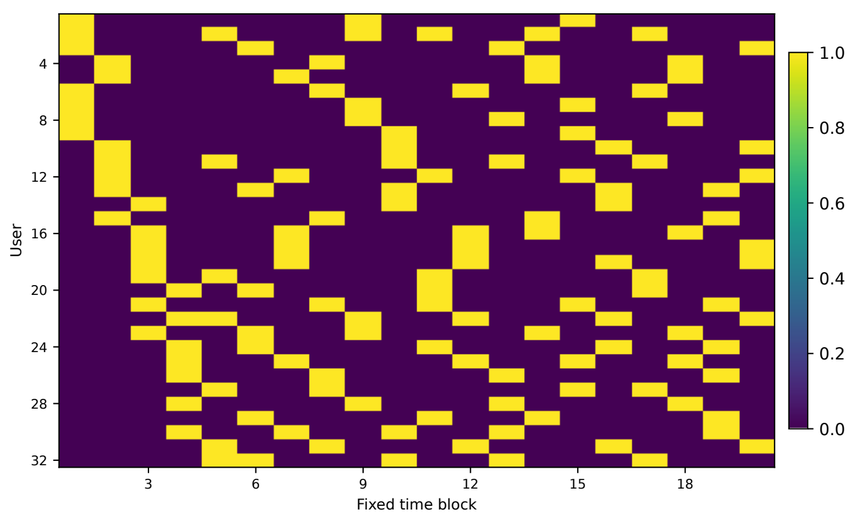
SB: adapt



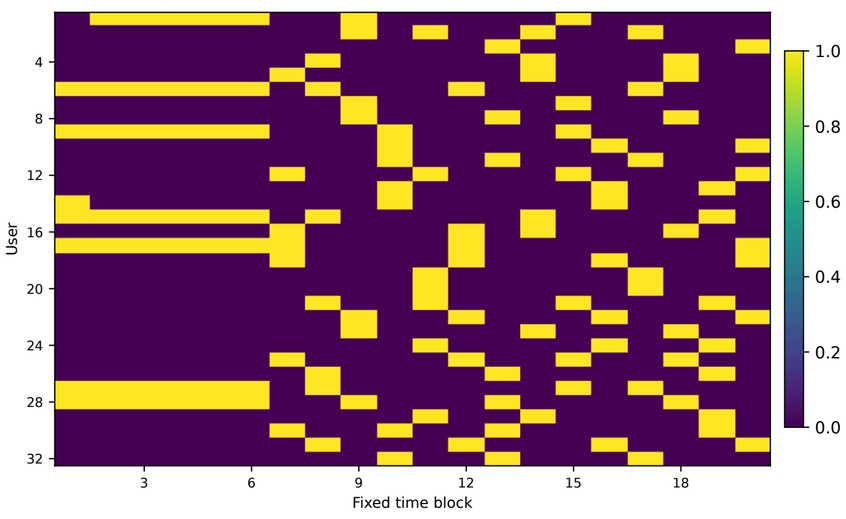
SB: light



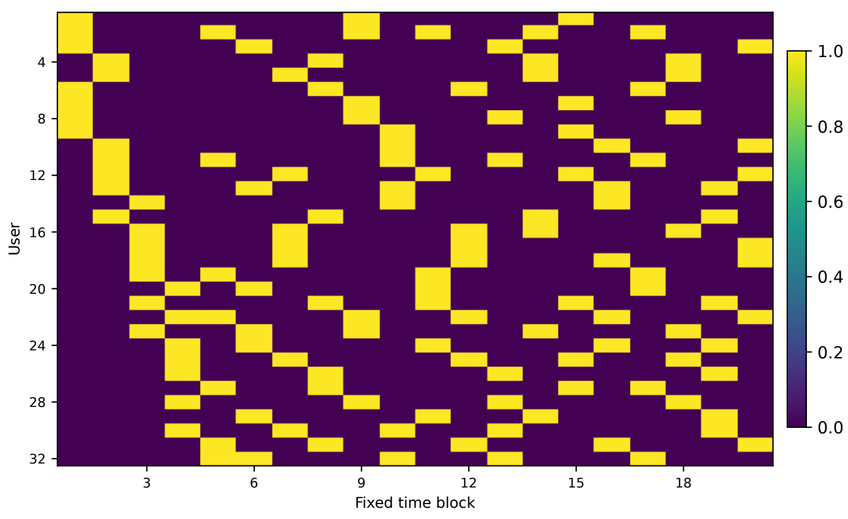
Price+Local



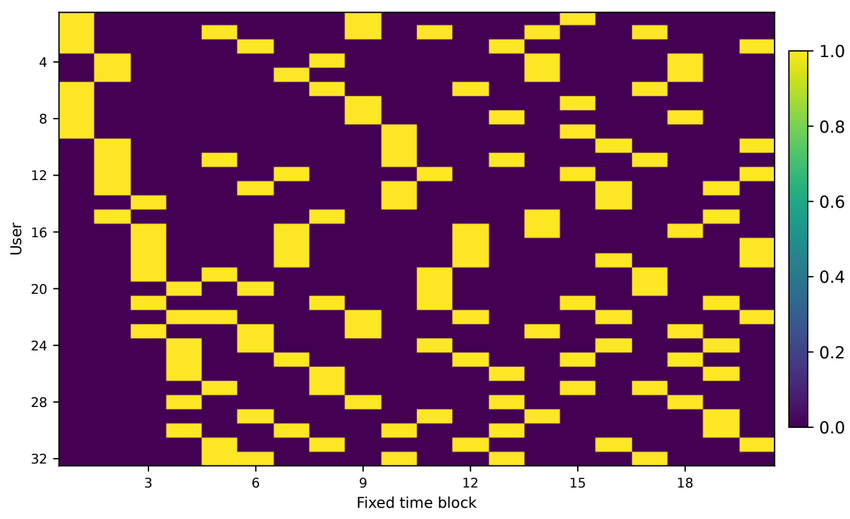
Local: adapt



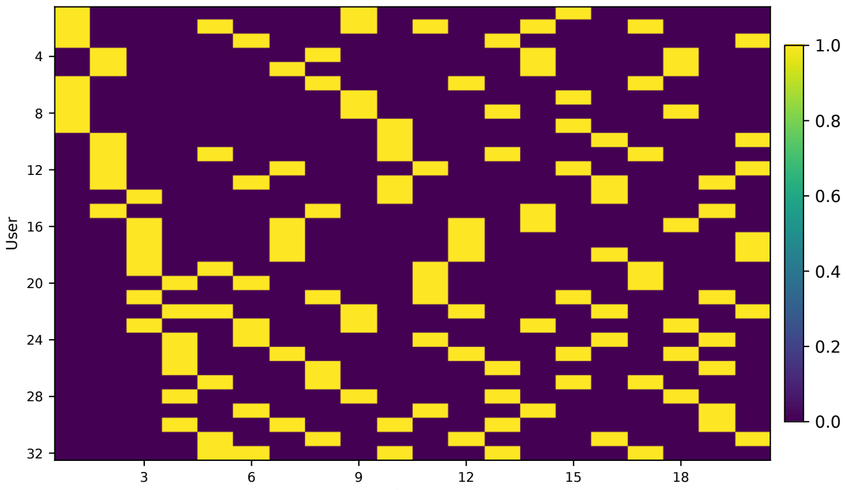
SB: zero



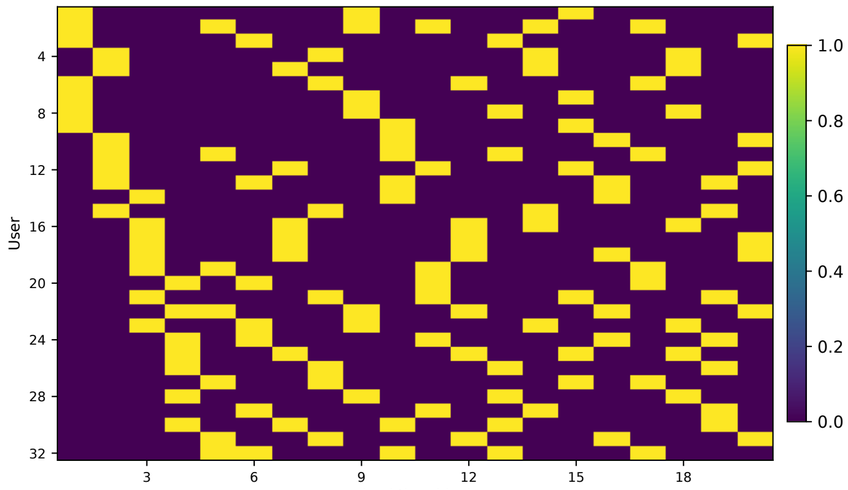
SB: minimax



SB: current

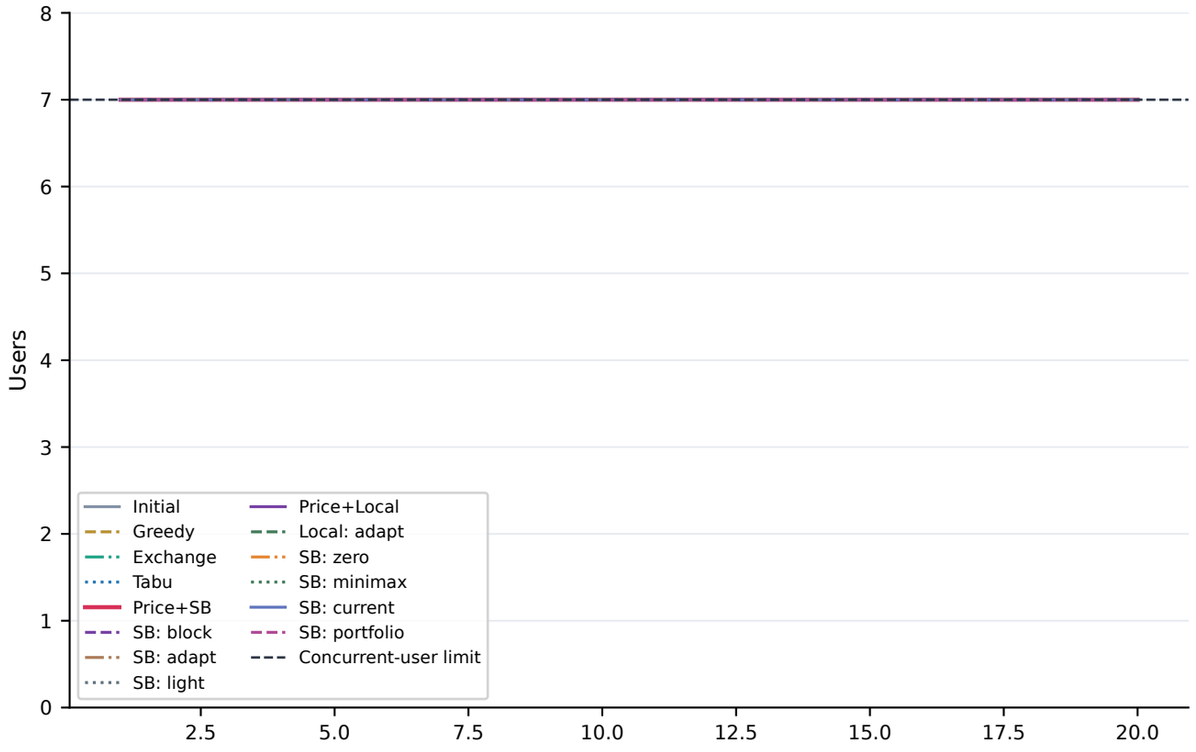


SB: portfolio

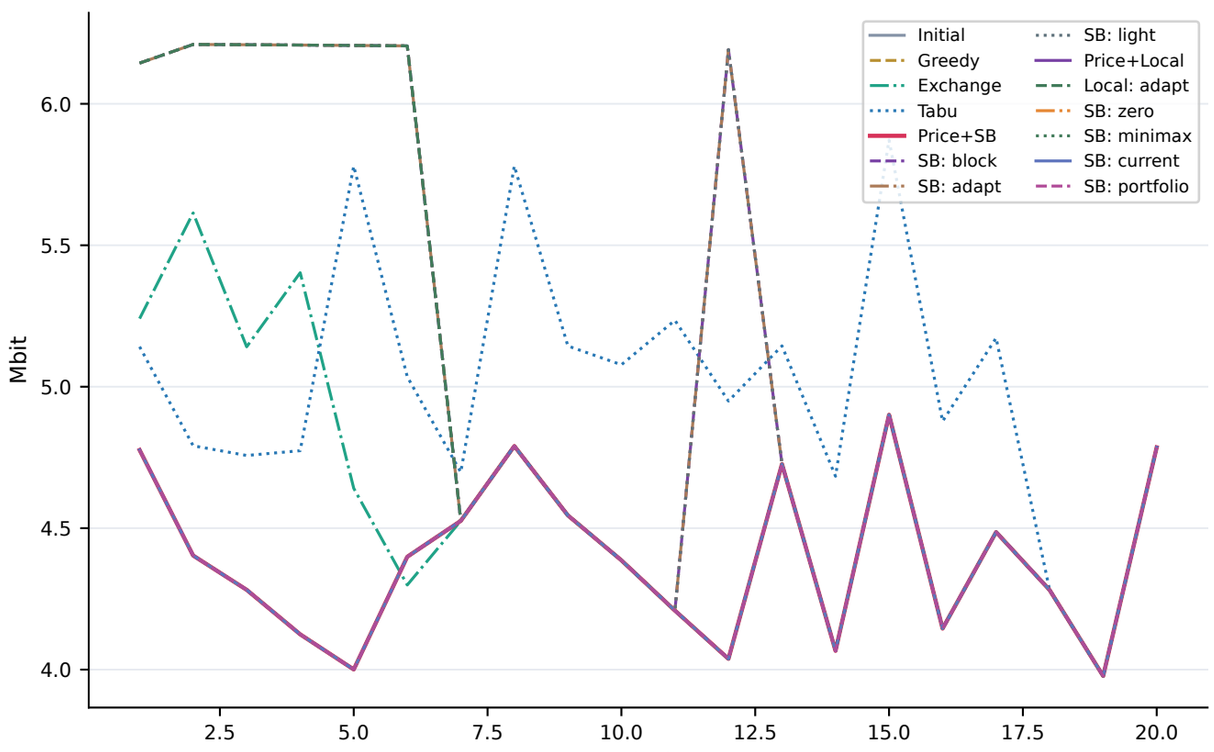


custom | first requested seed 71669974 | certified period service

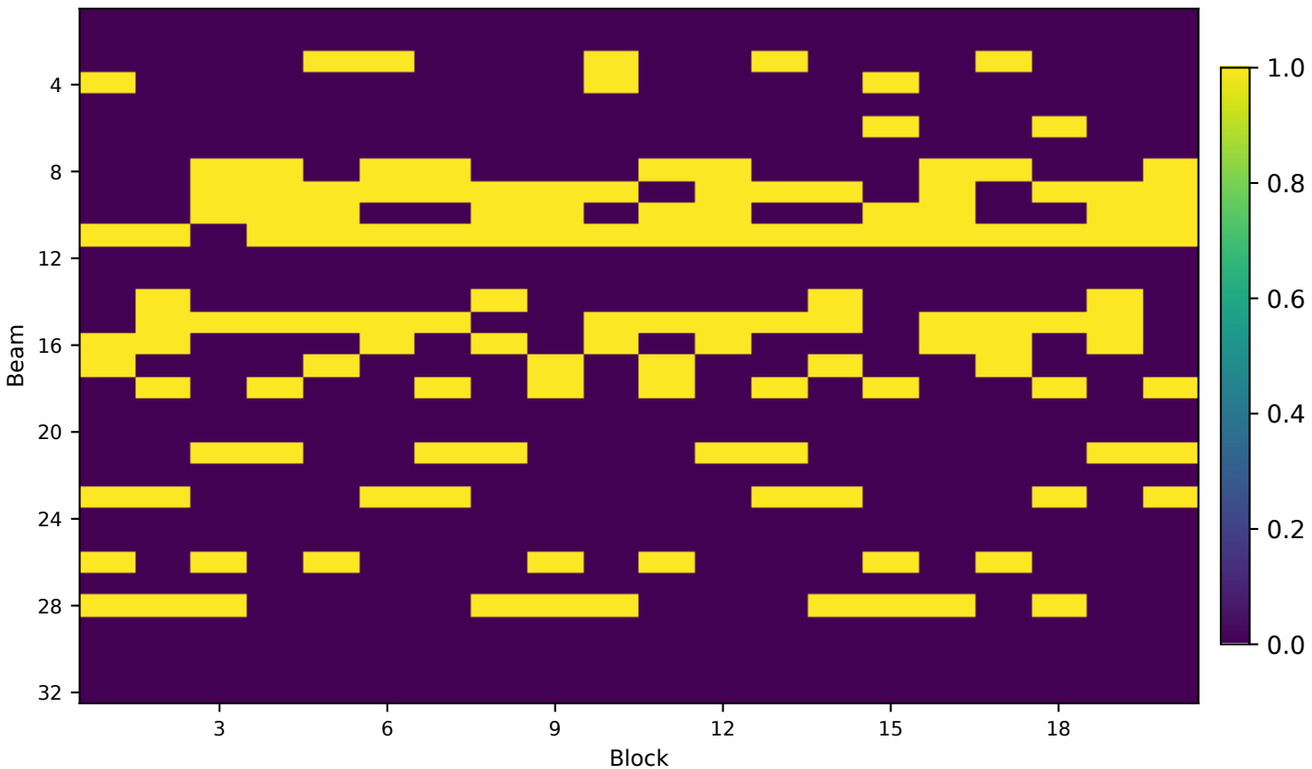
Concurrent users per fixed block



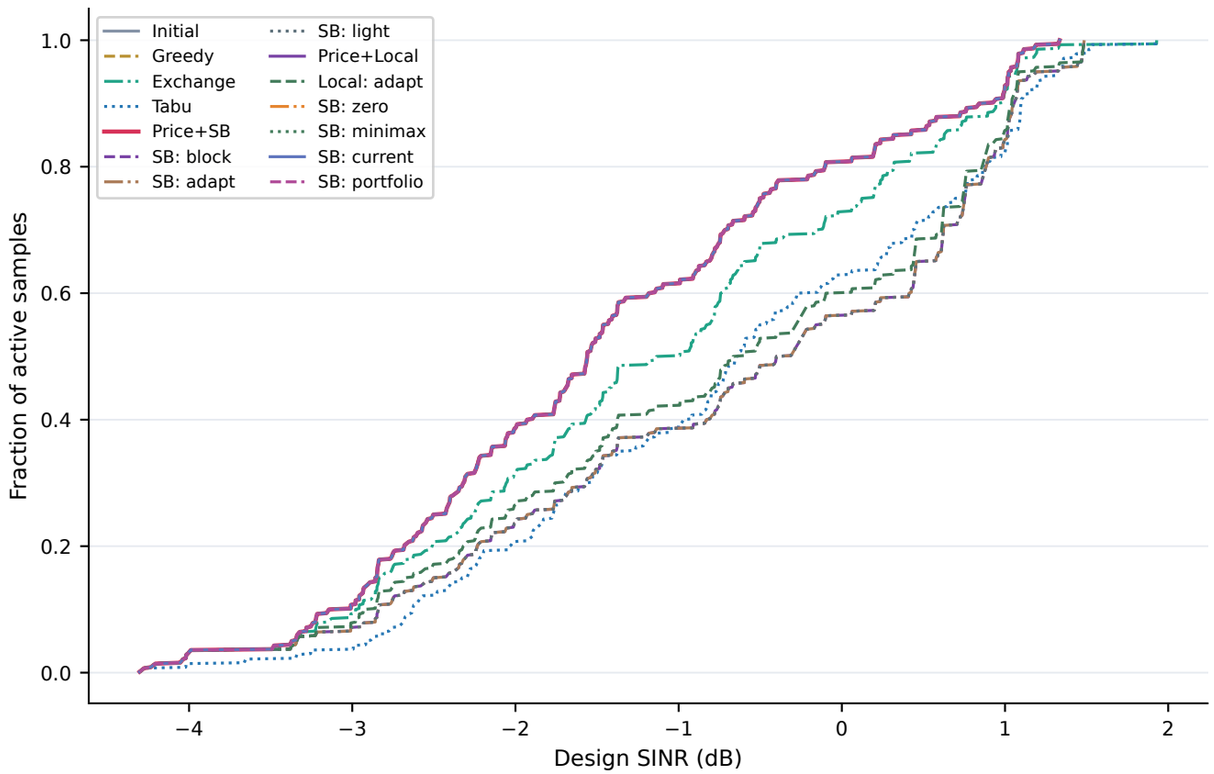
Additive service before period risk margin



Price+SB use of the fixed beam codebook

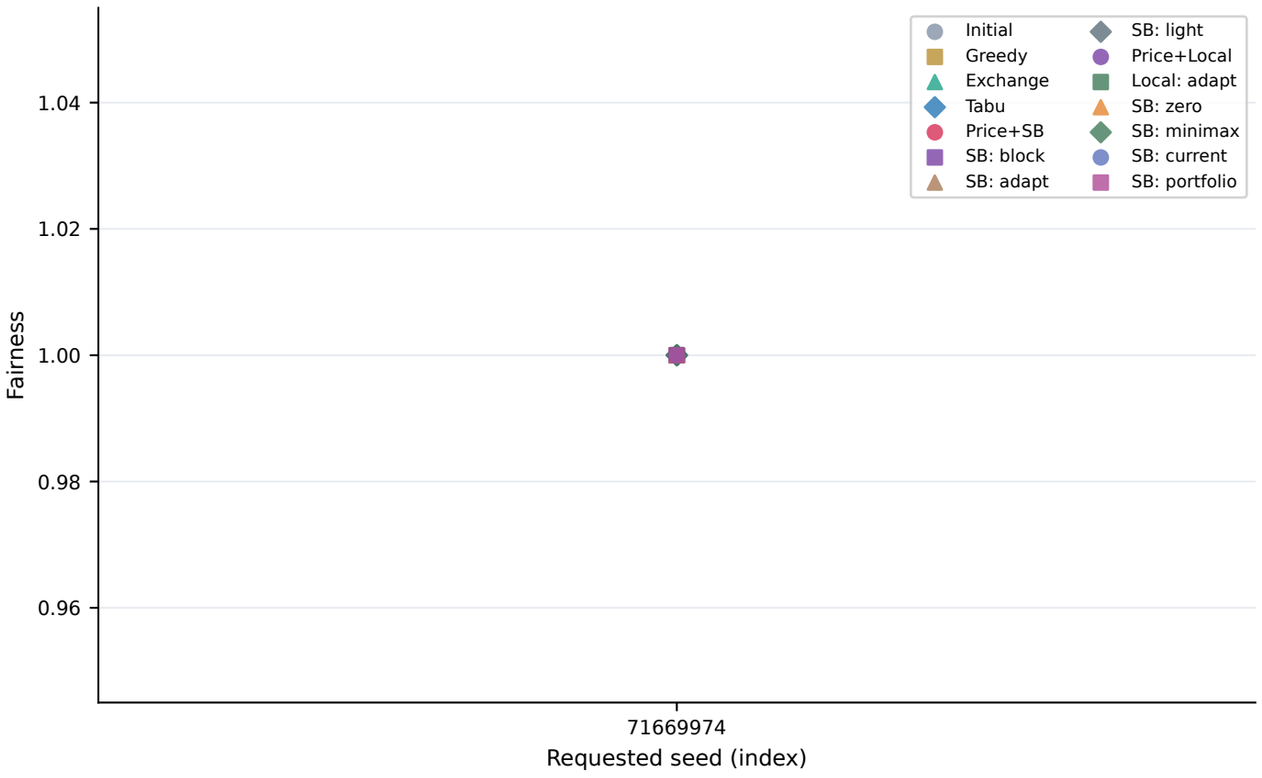


Design SINR CDF for active data samples

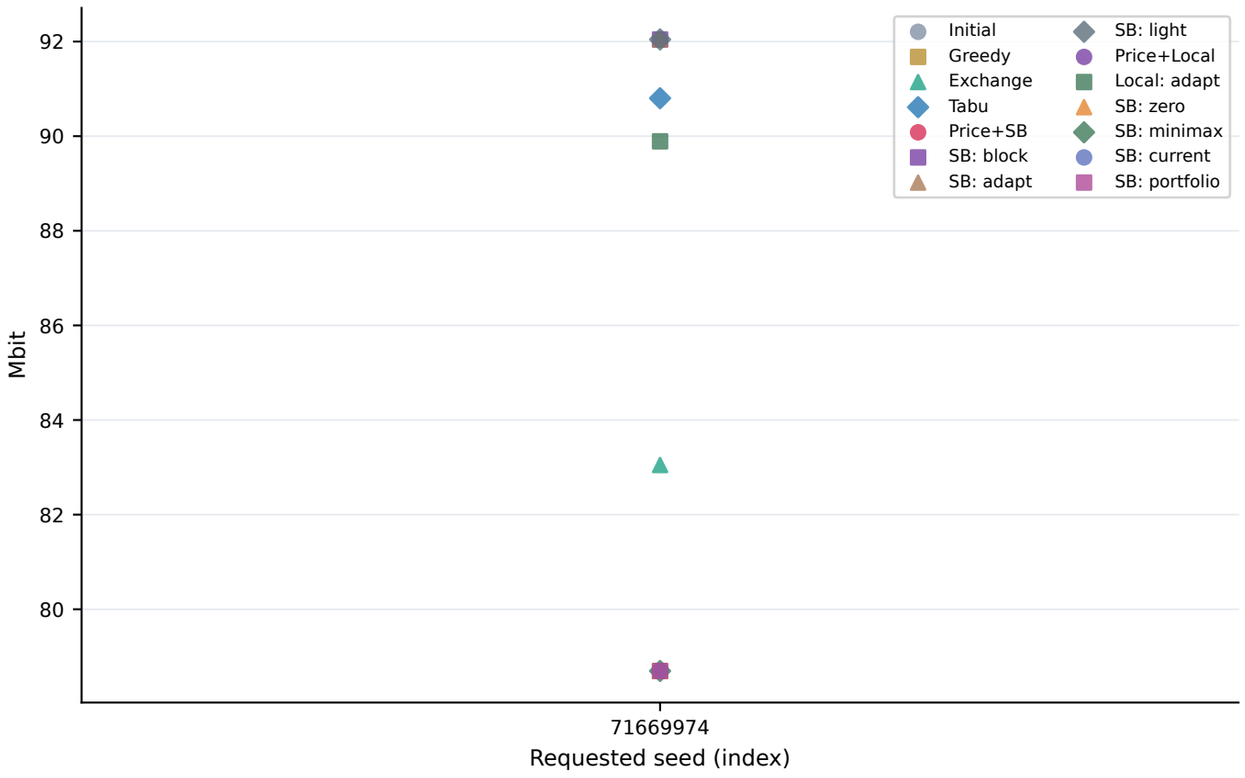


custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

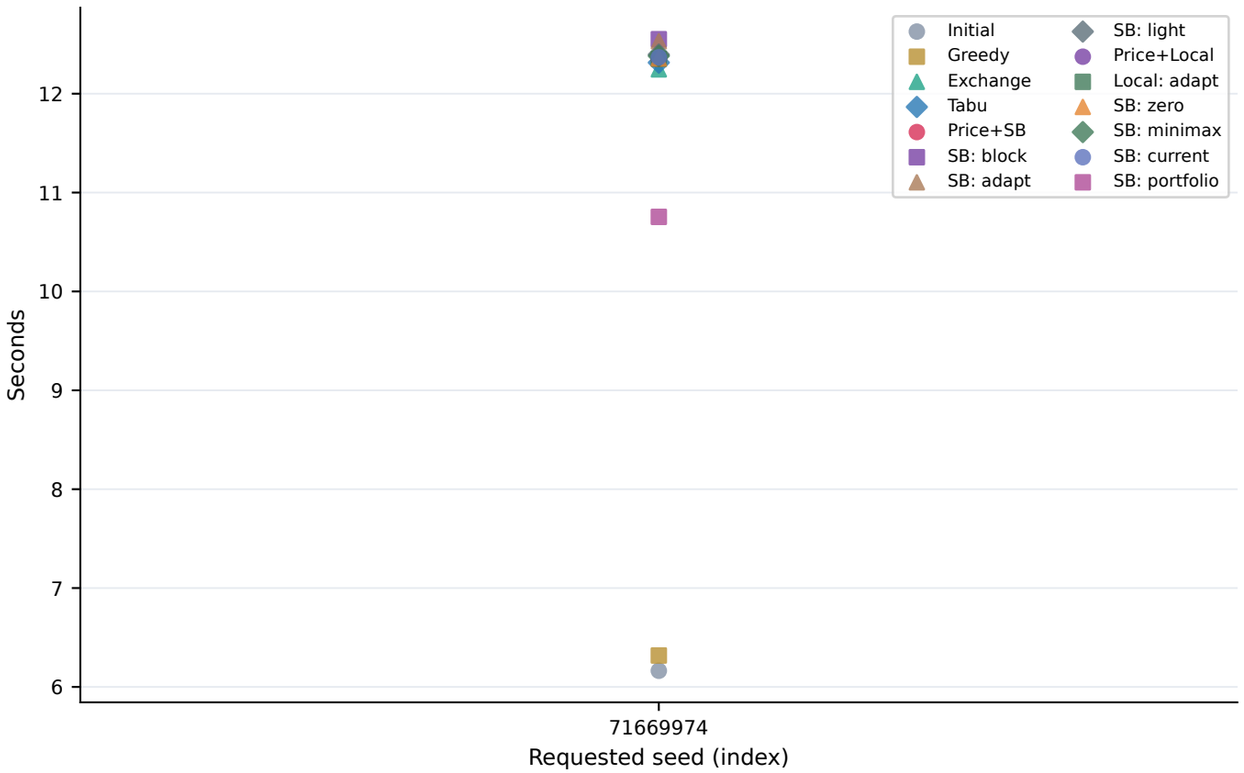
Fairness for every seed



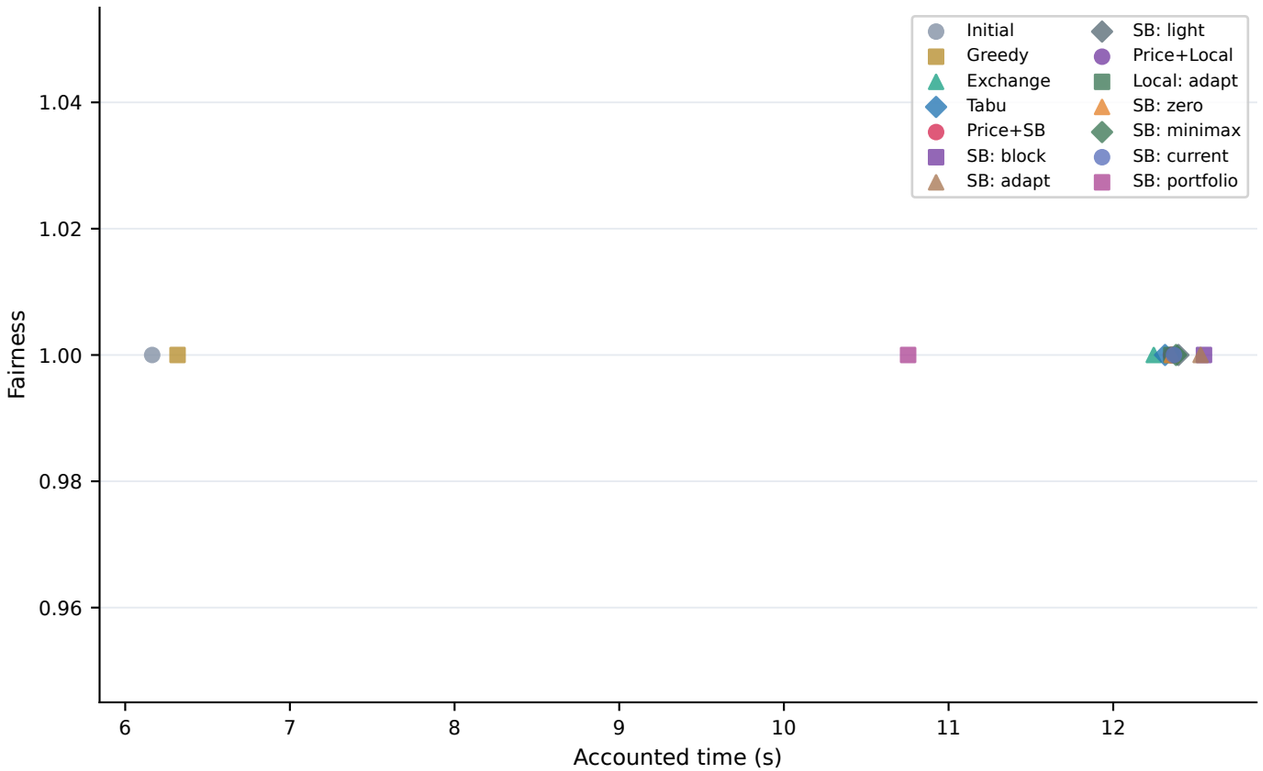
Throughput for every seed



Time for every seed

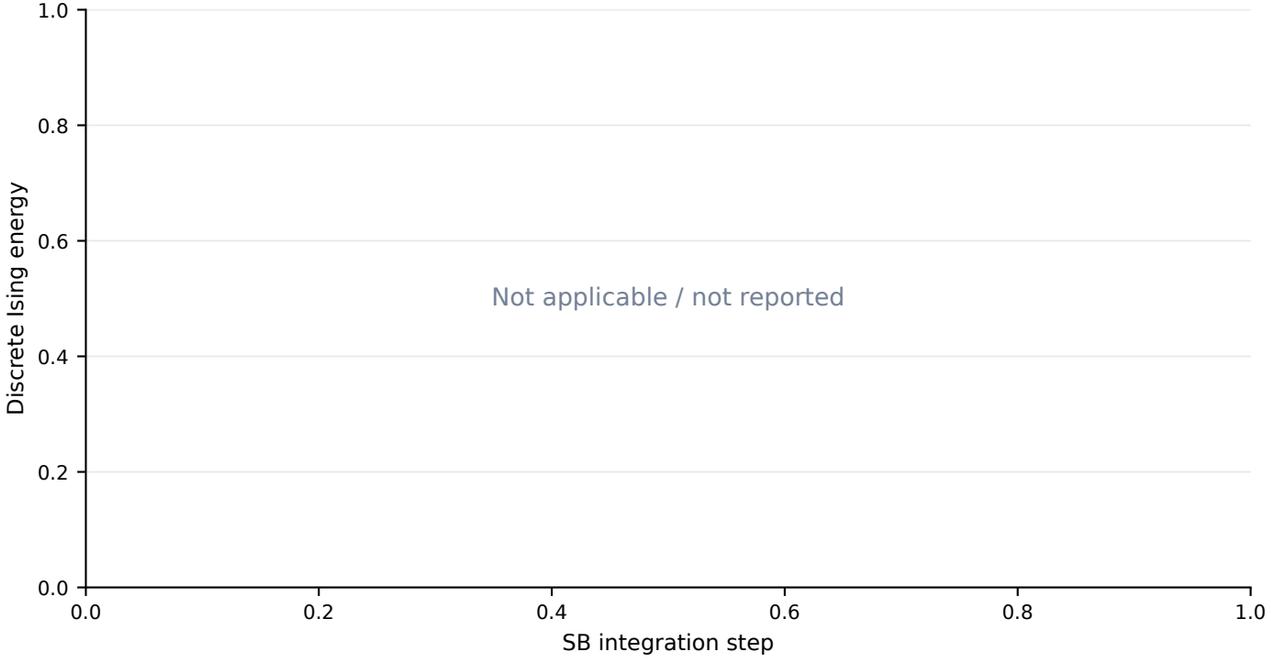


Quality versus time (every completed run)



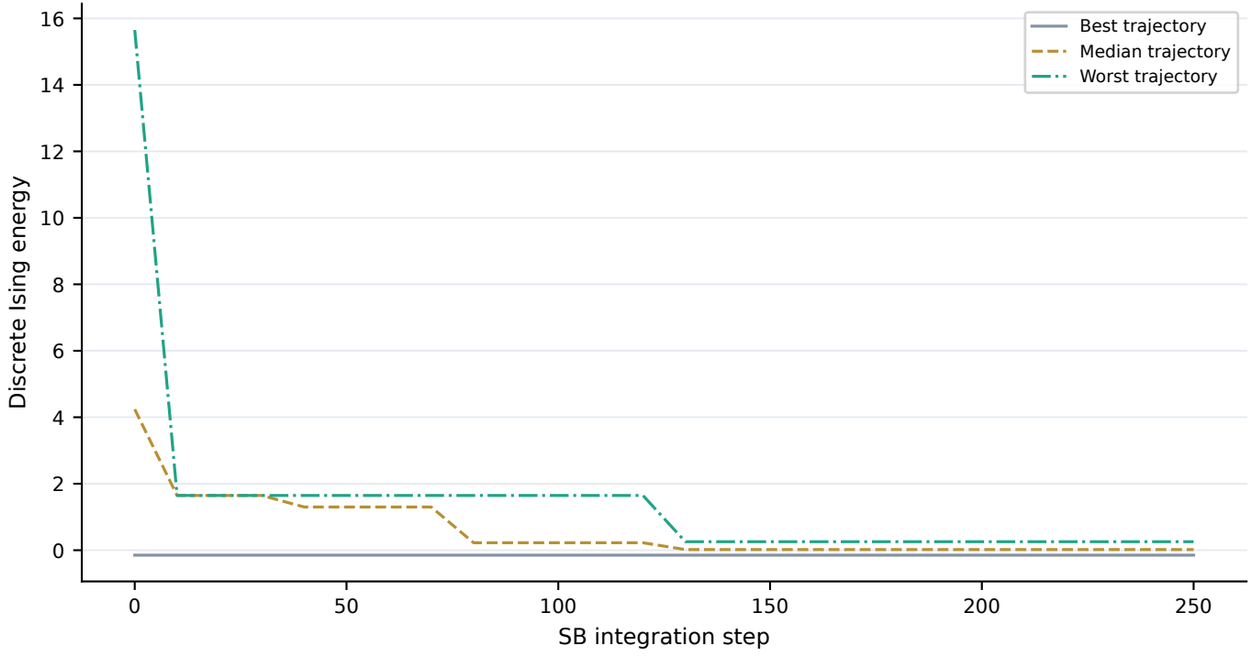
custom | first requested seed 71669974 | certified period service | first Price+SB Ising call per phase; before hard repair

Phase 1: fixed Ising call, best-so-far energy



Different phases solve different Ising problems; their energy values are not comparable.

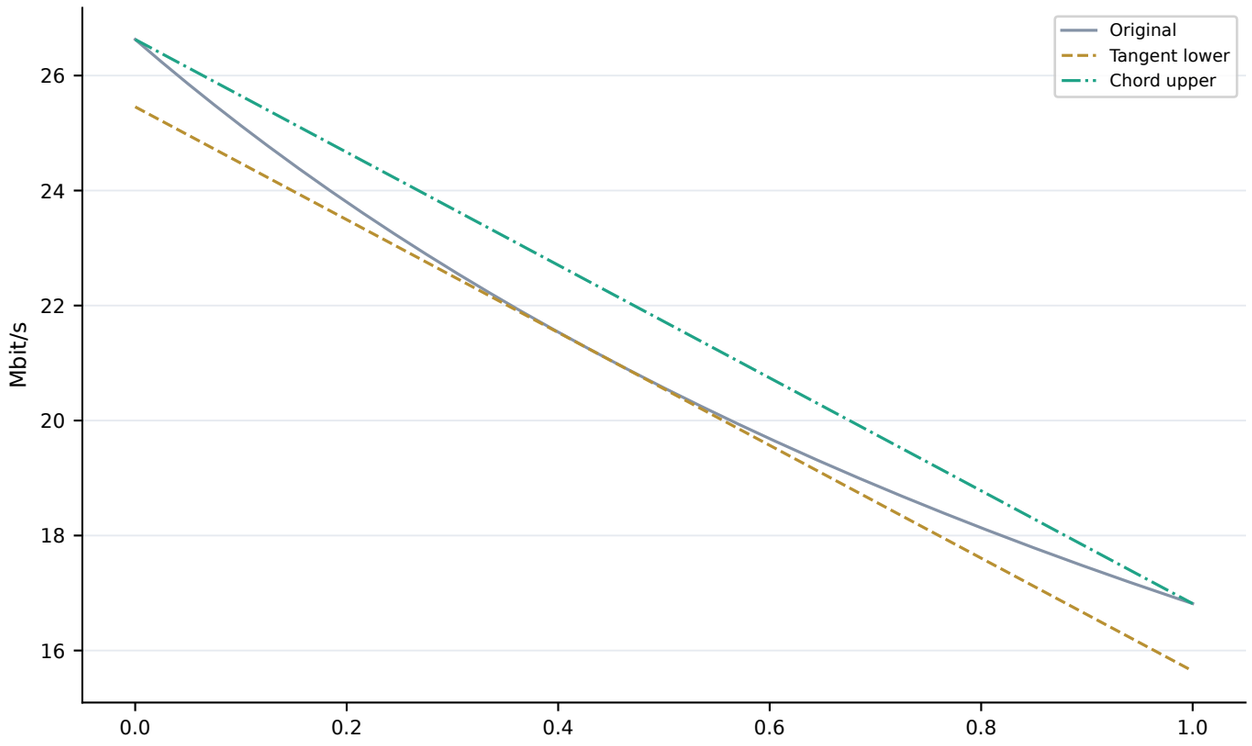
Phase 2: resource 1, price round 1, 16 trajectories



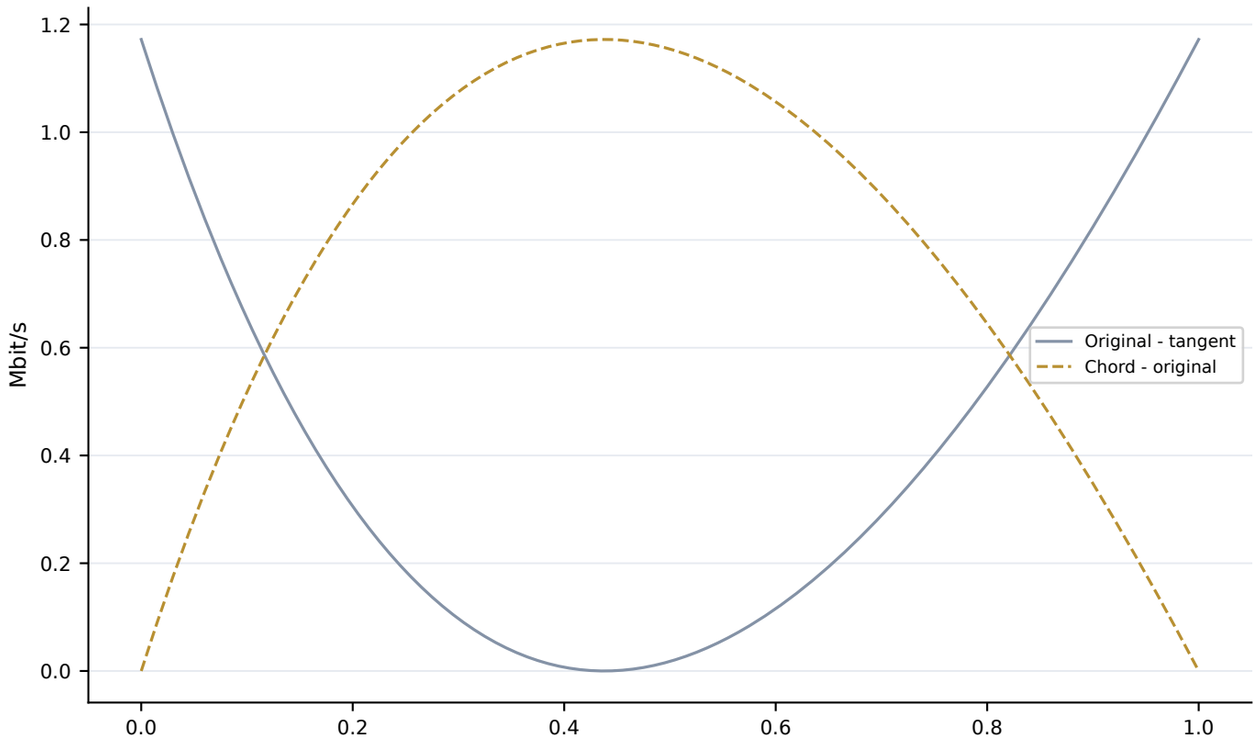
Different phases solve different Ising problems; their energy values are not comparable.

custom | first requested seed 71669974 | certified period service

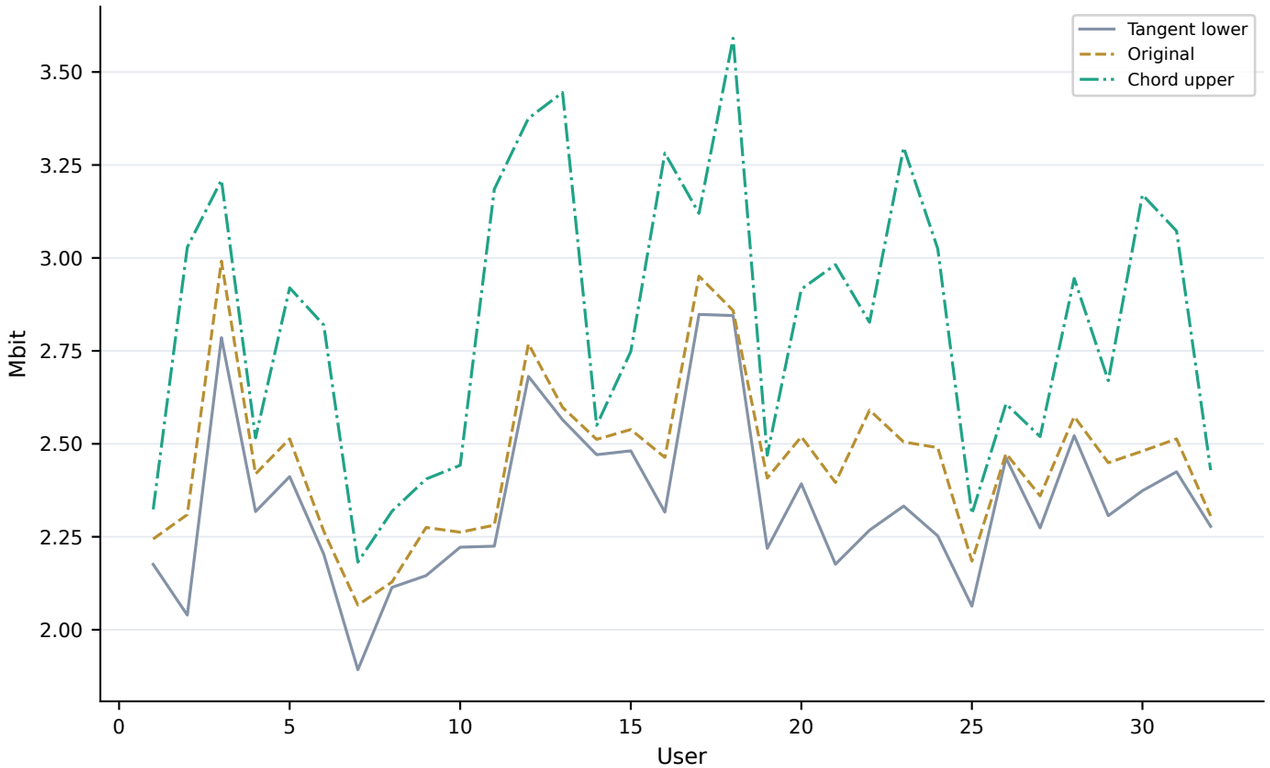
Rate bounds, first nonzero interval: user 1, sample 1



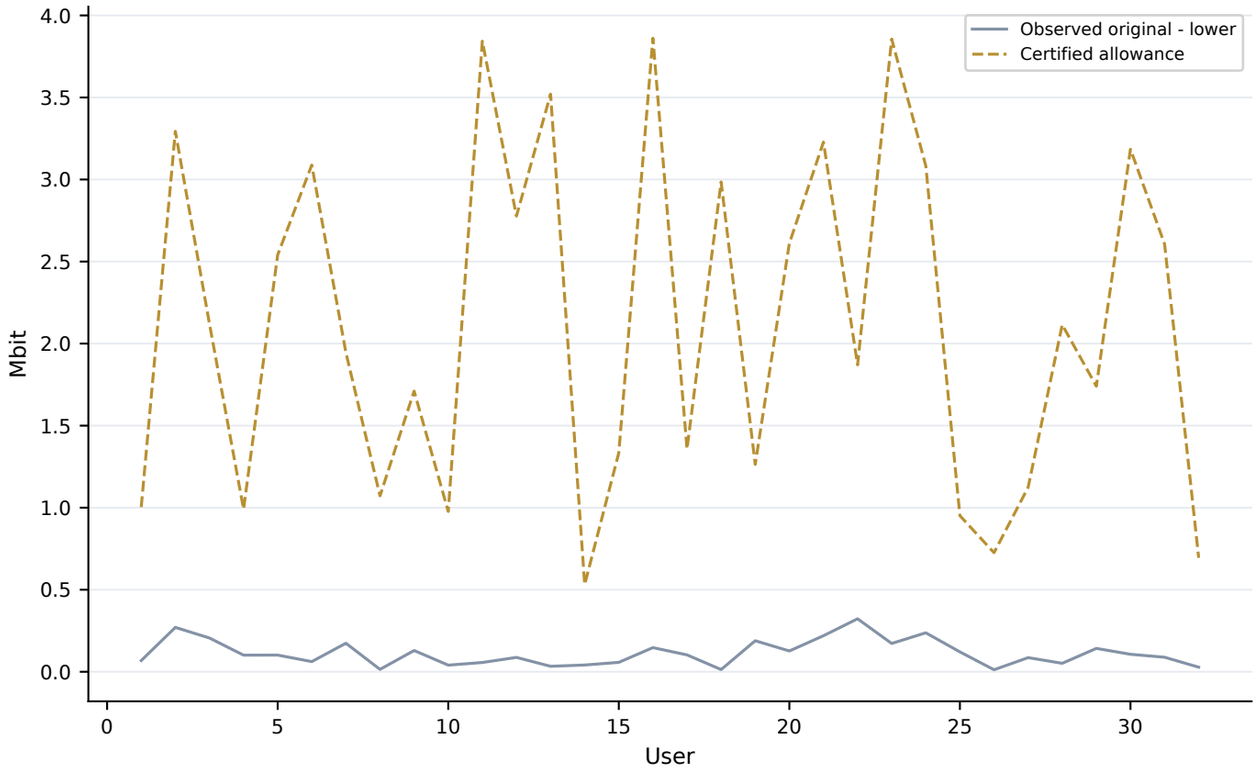
Rate approximation error over legal interval



Price+SB final schedule: fixed minimax service bounds

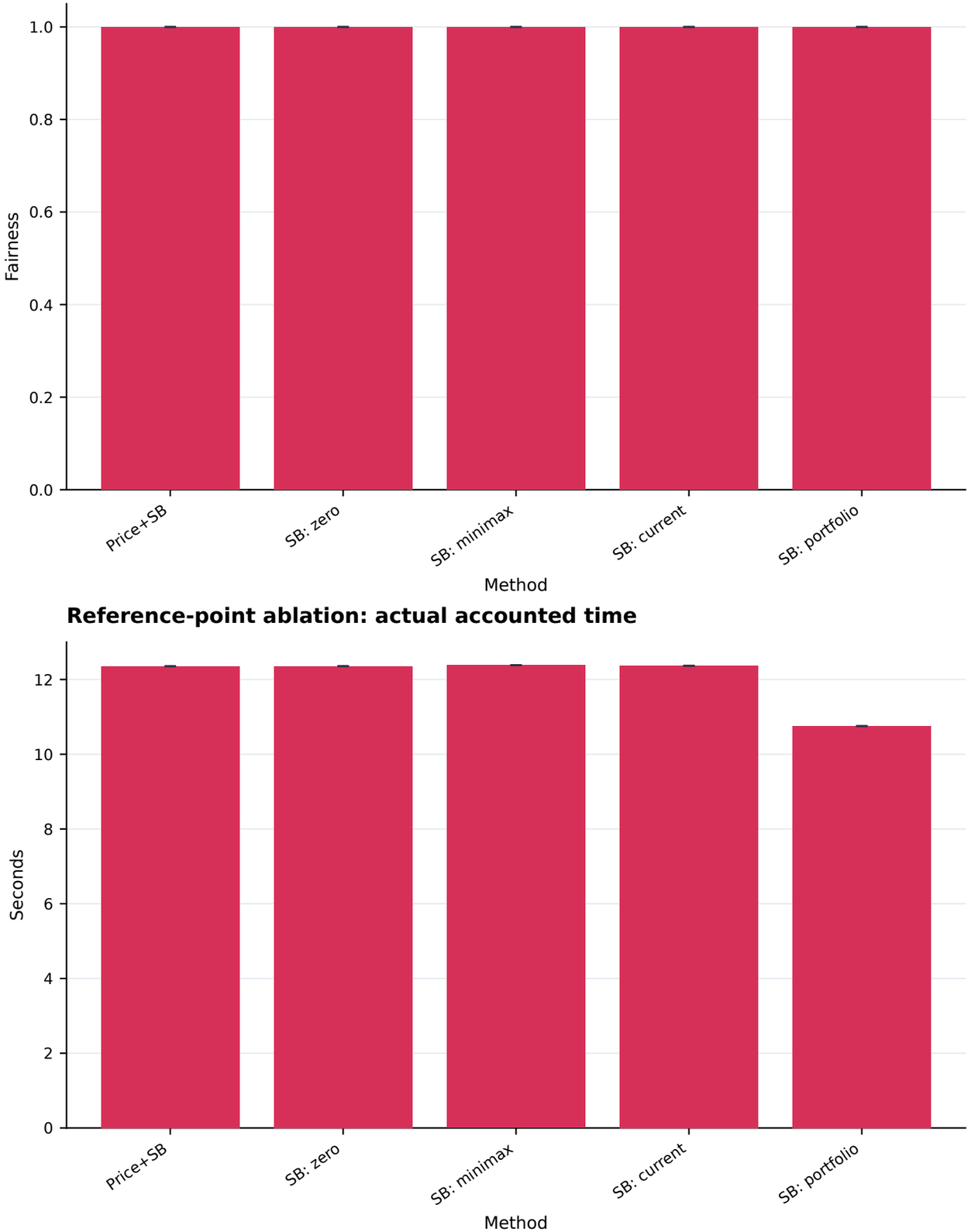


Observed lower error and certified error allowance

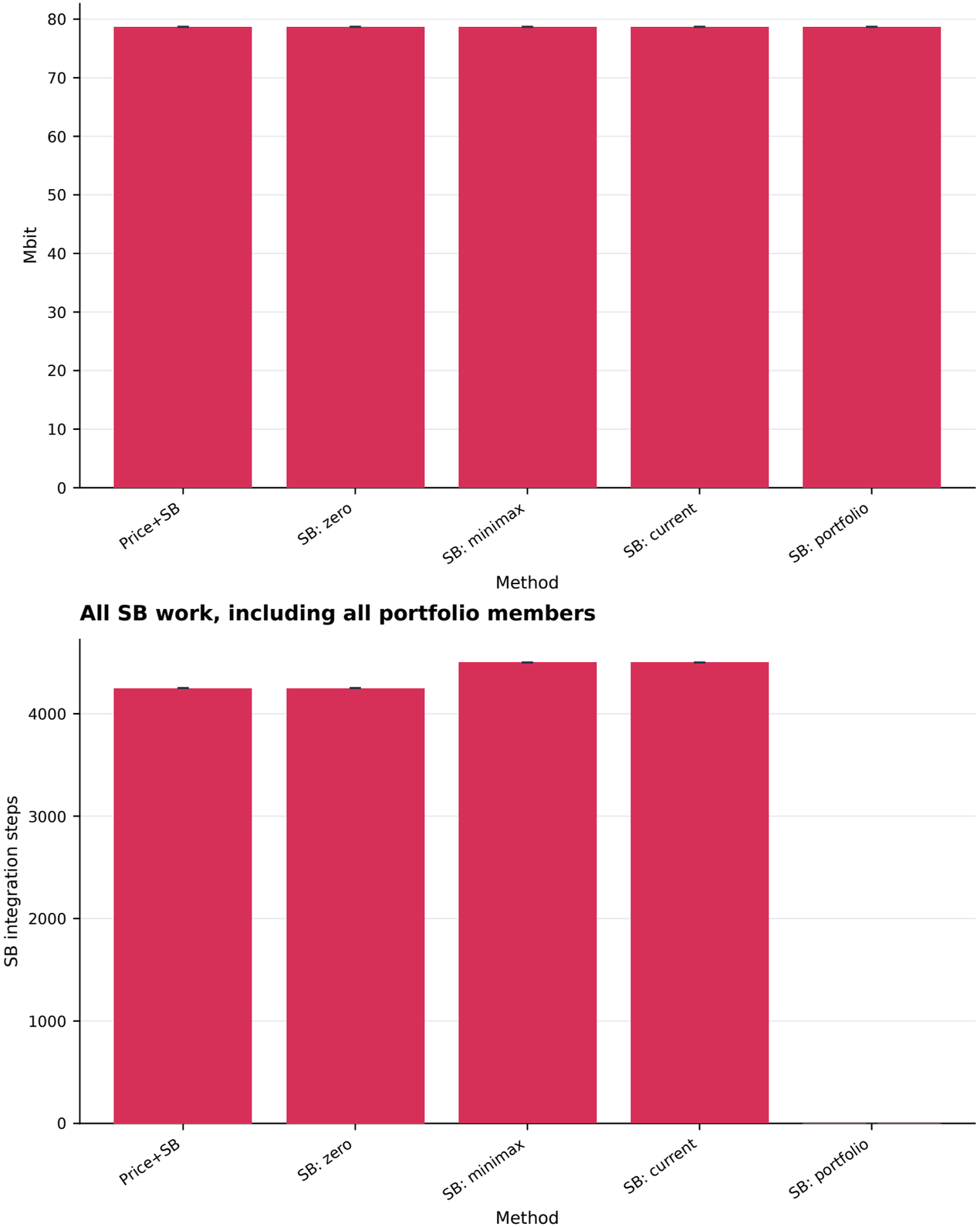


custom | K=32, M=20, L=7 | median and min-max across 1 seeds | certified period service

Reference-point ablation: original fairness



Reference-point ablation: original throughput



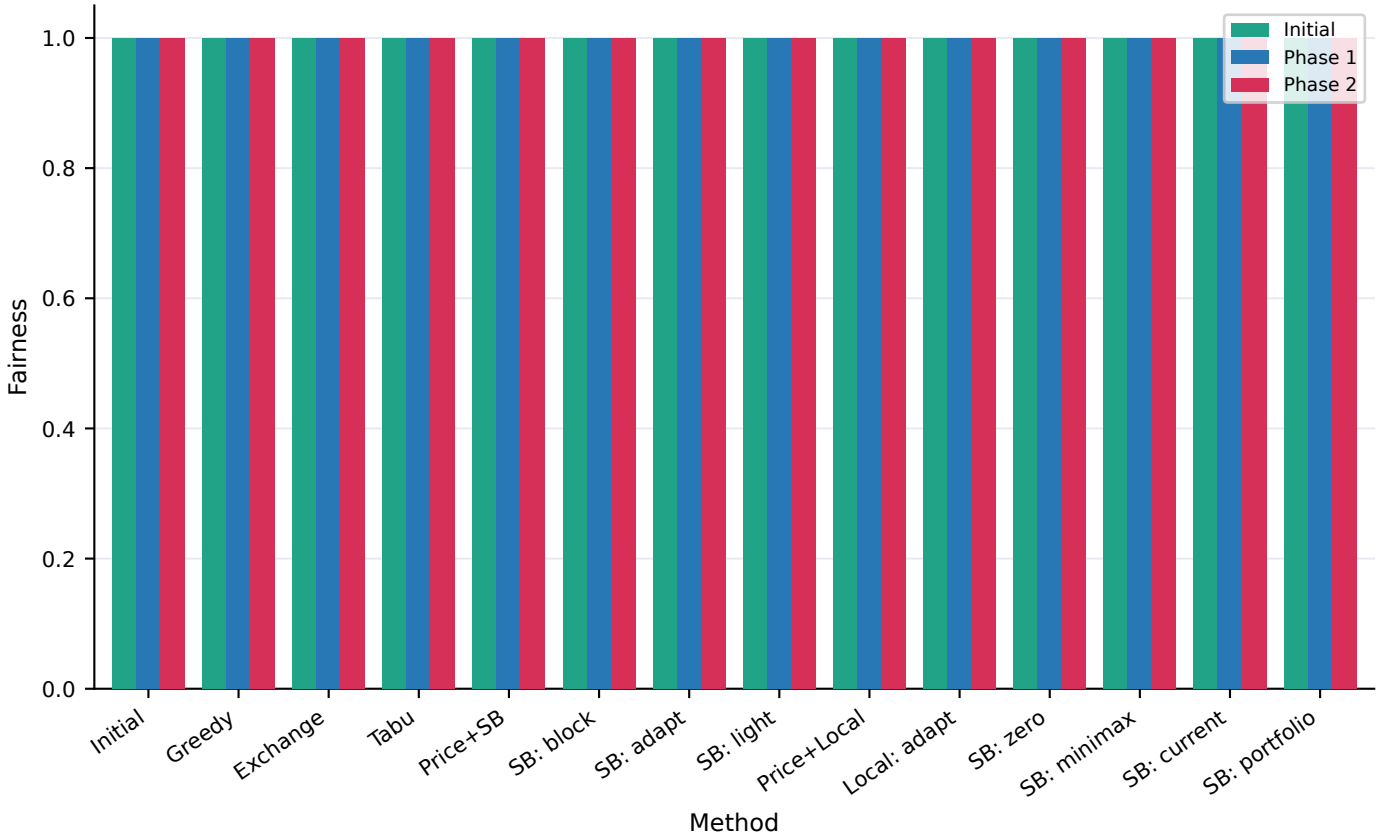
Reference-point ablation: actual accounted time

All SB work, including all portfolio members


custom | first requested seed 71669974 | certified period service
Method execution status across seeds
Initial
Greedy
Exchange
Tabu
Price+SB
SB: block
SB: adapt
SB: light
Price+Local
Local: adapt
SB: zero
SB: minimax
SB: current
SB: portfolio
Method
71669974
Seed
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Physical availability (all candidates, before scheduling)
User
4
8
12
16
20
24
28
32
3
6
9
12
15
18
Block
1.0
0.8
0.6
0.4
0.2
0.0
0=unavailable, 1=budget exhausted before search, 2=failed, 3=completed (may stop at time limit)

custom | first requested seed 71669974 | certified period service

Fairness at common initialization and each saved phase



Throughput at each phase (fairness is prioritized)

